

User Manual of WenJin Mincho

v2.011 2026.05(en)

GitHub: [takushun-wu/WenJinMincho](https://github.com/takushun-wu/WenJinMincho)

NOTE This user manual will be revised from time to time. This electronic document or its printed version may not reflect the latest updates. Users are advised to obtain the most current version of this manual.

Preface

Thank you for choosing WenJin Mincho. Unknowingly, the second major version of this project has arrived. Since its official release on January 3, 2025, the project has received over 400 stars on GitHub, and related posts on Zhihu and Rednote have accumulated over 20,000 views. It can be said that it has gained the support and recognition of a certain number of users. I would like to express my sincere gratitude here.

My initial motivation for developing WenJin Mincho was to satisfy the display of Unicode Han ideographs on my own computer devices. The predecessor of WenJin Mincho, SuperHan, was created between 2023 and 2024 based on Hanazono Mincho, Source Han Serif, and TW-Sung. At that time, the fonts that came with Windows (SimSun, MingLiU) could not achieve full Unicode Han ideograph coverage. At that time, there were almost no free commercial or open-source Song (Mincho) typefaces with full Unicode Han ideograph coverage mainly based on Mainland Chinese glyphs. The fonts available for free that displayed full Unicode Han ideograph coverage mainly included: TH-Tshyn fonts, Hanazono Mincho (predecessor of Jigmo), and Zhonghuasong(中华书局宋体). These fonts also failed to display ideographs with G (Mainland China) source according to the G source style (TH-Tshyn's Extension B, Extension E and later parts, the vast majority of ideographs in Hanazono Mincho, and Zhonghuasong's Extension C and later parts), which also brought some inconvenience. Therefore, I used data from GlyphWiki to organize and produce ideographs from G source and other sources.

Later, I learned about the Ideographic Variation Database (IVD), where adding an invisible Variation Selector (VS) character after a Han ideograph can achieve the display of variant ideographs. Considering that there are extremely few fonts that fully support IVD (not just the Adobe-Japan1 variant set contained in common Japanese body text fonts), I added IVD support to the SuperHan font and finally publicly released the SuperHan font with IVD support in March 2024.

After the release, it received some response. Subsequently, the higher quality SuperHan Next project was officially established, which is the current WenJin Mincho. In WenJin Mincho, I added support for multiple features such as IPA, romanization and Zhuyin systems, multiple languages, and glyph switching. At the same time, I also remade the glyphs that originally only had 256ppem TrueType curves into higher quality 1000ppem PostScript curve glyphs, corrected a large number of glyph errors, and also used the AFDKO suite used by Adobe for font development, making the development toolchain more complete than before. Finally, the first preview version was released in July 2024, with Unicode support version 15.1.

After half a year of iteration, WenJin Mincho has ushered in the time for the official release,

which is also the most anticipated moment. Overall, it has also received a very good response, which has brought great confidence to my subsequent font development. This is also a major source of motivation for my subsequent version iterations.

Finally, thank you again for your choice and recognition of WenJin Mincho, and I hope it will be helpful to your research and other work.

takushun-wu(@TakushunWu)

Introduction and Basic Information

WenJin Mincho is a large Han-ideograph character set Mincho(Songti) font suite developed with Source Han Serif and its derivatives, using GlyphWiki data and Han ideographs generated by a modified version of the Kage engine to achieve full coverage of Unicode-defined Han ideographs and Han ideographs registered in the Ideograph Variant Database, with Unicode Han ideographs are mainly rendered in mainland China glyph standards. The font family adopts *SIL Open Font License*, which is free for commercial use by any individual or organization.

There are relatively few free commercial and even open source large Han-ideograph character sets, and almost none of the free commercial large Han-ideograph character sets for Mincho(Songti) are based on the mainland China glyph standards. Therefore, if you are looking for a free commercially available large Han-ideograph character set font, WenJin Mincho is a good choice.

In addition, WenJin Mincho also supports a variety of Chinese romanization(or phonetic notations) and Zhuyin systems, and is equipped with a variety of OpenType features that can be freely invoked, including glyph standard switching, support for Chinese romanization(or phonetic notation) and Zhuyin variants, support for Zhuyin symbols in vertical typesetting, correct positioning of combining characters, and punctuation extrusion.

Before using the WenJin Mincho font suite, **it is important that you read the User's Manual in order to fully utilize the features offered by this font set and to avoid potential problems.** The OpenType features contained in this font are not well supported by all software. To display the best effect as much as possible, this manual is compiled using X_YLaTeX.

This font project will be updated irregularly, including function additions, improvements, and bug fixes. To minimize problems during use, it is recommended that users update in time. At the same time, this user manual will also be revised with version updates.

⚠ Important Notice

I DO NOT have an account on the GitCode platform under “Chongqing Open Source Co-creation (重庆开源共创)” or similar platforms, and have **NEVER** actively or authorized any third party to create repositories on GitCode or similar platforms. It is **STRICTLY FORBIDDEN** for code hosting platforms such as GitCode to privately move this font and the associated SuperHan font project without the written permission of the author, otherwise the author will pursue legal responsibility. The author is **NOT RESPONSIBLE** for any damage caused to third parties such as code hosting platforms due to the above actions. It is **STRONGLY RECOMMENDED** that users download from the official channels provided by the author.

Project Address, Changelog and Known Issues

1. Project Address
<https://github.com/takushun-wu/WenJinMincho>
2. Changelog (Since v2.000)
<https://github.com/takushun-wu/WenJinMincho/blob/main/CHANGELOG-en.md>
3. Known Issues Collection (Tencent Docs, in Chinese)
<https://docs.qq.com/doc/DRkpXRkx2S214QmNV>

Contact Information

If you encounter problems when using WenJin Mincho, please read the main text of this user manual and the *FAQ* chapter (page 45), the known issues collection, and the issue sections of the GitHub and Gitee repositories in detail to avoid duplicate and invalid questions, otherwise the author has the right not to deal with them.

I do not accept font customization requests for the time being. If you need business cooperation, it is recommended to contact via QQ.

-
- Repositories
<https://github.com/takushun-wu>
 - Email
junny_2016@qq.com
takushunwu@gmail.com
takushun-wu@outlook.com
 - Social Media Accounts
 QQ Group Chat (in Chinese) [Takufont 字体交流群 \(Takufont Exchange Group, 1053373658\)](#)
 QQ Private Account [takushun \(2833944613\)](#)
 Rednote(小红书) [takushun](#)
 X (Twitter) [@TakushunWu](#)

Contents

1	Quick Installation Guide	1
1.1	About Different Font Formats	1
1.2	Windows	2
1.3	macOS	3
1.4	Linux	3
1.5	iOS/iPadOS	4
1.6	WPS Office for Android/Harmony OS	5
2	Basic Functions	7
2.1	Unicode Han Ideographs Support	7
2.1.1	CJK Unified Ideographs and Compatibility Ideographs Support	7
2.1.2	Standardized Variants and Ideographic Variation Database Support	8
2.2	Support for Multiple Romanization and Zhuyin Systems	8
2.3	International Phonetic Alphabet (IPA) Support	10
2.4	Latin-Greek-Cyrillic (LGC) Character Support	10
2.5	Special Symbol Support	11
3	Introduction to OpenType Features	13
3.1	Han Ideograph Composition and Glyph Switching	13
3.1.1	Standard Glyph Switching	13
3.1.2	Ancient Book Standard Glyph Switching	14
3.1.3	Vietnamese Tone Mark Combination	15
3.2	OpenType Features for Romanization, Zhuyin, IPA Symbols	15
3.2.1	Hànyǔ Pīnyīn Handwriting Support	15
3.2.2	Superscript and Subscript Numbers	15
3.2.3	Free Switching of Zhuyin Tone Mark Forms	15
3.2.4	Support for Some Zhuyin Symbols Not Included in Unicode	16
3.2.5	Zhuyin Vertical Layout Support	17
3.2.6	Historical Forms of Zhuyin Symbols	17
3.2.7	Correct Positioning of Combining Characters	17
3.2.8	Form Conversion of Four-Corner Tone Marks	18
3.3	Non-CJK Language Related OpenType Features	18
3.3.1	Non-CJK Punctuation Display	18
3.3.2	Language-Dependent Contextual Substitution	19
3.3.3	Italic Character Substitution	20
3.3.4	Slashed Zero	20

3.3.5	Fraction Related OpenType Features	21
3.3.6	Ordinal Form Substitution	21
3.3.7	Tabular Figures Support	21
3.4	Other OpenType Features Introduction	21
3.4.1	Punctuation Kerning Support	21
3.4.2	Centered Punctuation Display	22
3.4.3	Vertical Punctuation Display	22
3.4.4	Full-width/Half-width/Proportional Character Switching	22
3.4.5	Discretionary Ligatures	23
3.4.6	Squared Numbers and Letters	23
3.4.7	Alternate Annotation Forms for Ordinal Characters	23
4	OpenType Feature Usage Guide	25
4.1	HTML/CSS	25
4.2	LaTeX	28
4.2.1	General Settings	28
4.2.2	X _Y LaTeX	30
4.2.3	LuaLaTeX	30
4.3	Adobe InDesign	32
4.4	Other Software	33
4.4.1	Other Adobe Design Software (such as Photoshop, Illustrator, etc.)	33
4.4.2	Microsoft Word	33
4.4.3	Microsoft PowerPoint	34
4.4.4	macOS iWork Suite (Pages, Numbers, Keynote), TextEdit, etc.	34
4.5	Alternative Solutions for Applying OpenType Features in Environments that Do Not Support Them	35
5	System Fallback Font Installation Guide	37
5.1	Installing as a Windows System Fallback Font	37
5.2	Installing as a Fallback Font for Android System (or AOSP-based Systems)	38
6	Important Notes	41
7	Font License	43
7.1	You are permitted to:	43
7.2	You are strictly prohibited from:	43
7.3	Disclaimers and Declarations	44
8	Frequently Asked Questions	45
A	SIL Open Font License	49
B	Donation Support	53

Chapter 1

Quick Installation Guide

Thank you for choosing WenJin Mincho. Before using WenJin Mincho, you generally need to install the font on your device before it can be used by applications. The following will describe the installation methods for common platforms.

1.1 About Different Font Formats

Caution!

Starting from February 16, 2026, the Compact Line-Spacing Version (C Version) and Windows GDI Compatible Version (W Version) will no longer be updated. The C Version and W Version released before this date will be the last versions. Users of C Version and W Version are urged to update and replace them with the standard version as soon as possible, try to use newer versions of software, and keep the font version up to date. The author apologizes for any inconvenience caused.

If you have difficulty replacing the standard version (such as incompatibility with new software), you can modify the standard version font file yourself to meet your needs. Note that the modified version still needs to comply with the license attached to the font (e.g., modified versions of OFL fonts must still choose the OFL license).

Regardless of the version, or the three font files and TTC integration package contained in each version, the font data and functions are basically the same. Users only need to choose one of the three font files or the TTC integration package to install.

OTF Font Files (PostScript Curves)

- `otf/WenJinMinchoP[0|2|3]-Regular.otf` Three separate OpenType font files, each containing ideographs for the corresponding plane, with the number representing the plane number.
- `ttc/WenJinMincho-OTF.ttc` Integration package containing the above three OTF fonts.

macOS/iOS/iPadOS/Linux/Android/BSD and other (Unix-like) systems are recommended to choose this version for installation; Windows users who frequently use Adobe software and $\LaTeX$ can also choose this version for installation.

TTF Font Files (TrueType Curves)

- `ttf/WenJinMinchoP[0|2|3]-Regular.ttf` Three separate TrueType font files, each containing ideographs for the corresponding plane, with the number representing the plane number.
- `ttc/WenJinMincho-TTF.ttc` Integration package containing the above three TTF fonts.

General Windows users and those planning to use it as a Windows fallback font are recommended to choose this version for installation. Windows and Office have the best compatibility with TTF version fonts.

Compact Line-Spacing Version TTF Font Files (TrueType Curves, Ended Support) ▲

- The font name and PostScript name of the compact line-spacing version are different from the original version, adding a letter “C” to distinguish.

Compared with the above versions, only the line spacing parameters of the font have changed.

Windows GDI Compatible Version TTF Font Files (TrueType Curves, Ended Support) ▲

- The font name and PostScript name of the Windows GDI compatible version are different from the original version, adding a letter “W” to distinguish.

If you use standard and compact line-spacing version TTF fonts on the Windows GDI environment in Windows systems (such as Word 2010, BabelStone Pad), you can find that quotation marks are displayed in non-CJK style, left quotation marks in Word 2010 are located on the left with a blank space, or some punctuation marks are displayed in non-CJK style (variable width, not Chinese full-width form). Only the Uniscribe shaping engine has the above problems; DirectWrite, HarfBuzz, etc. do not have this problem. Based on the compact line-spacing version, this version removes the automatic conversion of some punctuation marks to variable-width half-width forms when the text is in a non-CJK environment (default mapping is full-width form).

1.2 Windows

- 1 Right-click the font and click **Install** to install the font for the current user; click **Install for all users** (requires administrator privileges; Windows 11 may also need to click **Show more options** first) to install the font for all users (as a system font);
- 2 You can view the installed fonts in applications (some programs need to be restarted to view).

Note

- Some programs may need to install fonts for all users to recognize them correctly;
- To use as a system fallback font requires additional configuration (see Section 5.1, page 37),

and it is not effective in all scenarios;

- Path for installing fonts for the current user:
`<System Drive>:\Users\<Current User>\AppData\Local\Microsoft\Windows\Fonts`
- Path for installing fonts for all users (as system fonts): `<System Drive>:\Windows\Fonts`

1.3 macOS

- 1 Open the Font Book App and click the + (plus sign) in the menu bar;
- 2 In the pop-up dialog box, select the font file and click **Open** to complete the installation.

Note

- You can choose whether the font is available only to the current user or to all users by modifying the *Default Install Location*;
- After installation, the system automatically uses WenJin Mincho as the system fallback font, generally requiring no additional configuration (unless your computer also has other large Han ideographs set fonts installed);
- Path for installing fonts for the current user: `/Users/<Current User>/Library/Fonts/`
- Path for installing fonts for all users (as system fonts): `/Library/Fonts/`

Tip

If you have the Homebrew package manager installed on macOS, you can also install the OTC version of WenJin Mincho through Homebrew with the following command (macOS version must be 10.15 or above):

```
1 brew install --cask font-wenjin-mincho
```

1.4 Linux

- 1 Open the terminal (find in App library or enter `Ctrl+Alt+T` shortcut);
- 2 If installing fonts for the current user, please enter and press Enter:

```
1 mkdir ~/.fonts # If the folder already exists, no need to create it again
2 cp <font path> ~/.fonts
3 fc-cache -f -v
```

If installing fonts for all users (as system fonts), please enter and press Enter (requires administrator privileges):

```
1 sudo mkdir /usr/share/fonts/wenjin-mincho
2 sudo cp <font path> /usr/share/fonts/wenjin-mincho
3 sudo fc-cache -f -v
```

- 3 Wait for the above commands to complete, you can enter the `fc-list` command to view the installed fonts. If the font just installed exists, it means the font has been successfully installed and can be called by applications.

i Note

- If you encounter a command not found error when executing `fc-list` or `fc-cache` commands, please install the `fontconfig` package first (the specific method varies by distribution);
- The installation path for fonts installed by the current user may also be `~/.local/share/fonts`, subject to the actual situation of the distribution;
- It can also be installed through the font manager included in the GNOME or KDE desktop environment;
- After installation, the system automatically uses WenJin Mincho as the system fallback font, generally requiring no additional configuration (unless your computer also has other large Han ideographs set fonts installed). If necessary, please refer to the following link (in Chinese):

<https://szclsya.me/zh-cn/posts/fonts/linux-config-guide/>

💡 Tip

Arch Linux users can also install the OTF version of WenJin Mincho through the pacman package manager, with the following steps (the corresponding package is maintained by a third party):

- 1 Add the following content to the `/etc/pacman.conf` file (i.e., add the Arch Linux Chinese community repository software source, or its [mirror address](#)):

```
1 [archlinuxcn]
2 Server = https://repo.archlinuxcn.org/$arch
```

- 2 Enter the following command to import the PGP key:

```
1 sudo pacman -Sy && sudo pacman -S archlinuxcn-keyring
```

- 3 Enter the following command to install the OTF version of WenJin Mincho:

```
1 sudo pacman -Syu otf-wenjinmincho
```

1.5 iOS/iPadOS

- 1 Download and install **iFont** from the App Store, then open the App;
- 2 Click **Import** → **Import from files**, and select the font file in the pop-up file dialog box;
- 3 In the main menu of iFont, click **Imported**, select the font just added, click **INSTALL**, and then click **Continue** in the secondary menu;
- 4 A prompt about downloading the configuration profile from *localhost* will appear in the pop-up window. Click **Allow**, then a **Profile Downloaded** dialog box will pop up, click **Close**;
- 5 Return to the desktop, open **Settings**, the main settings page will display the **Profile Downloaded** option, click to enter;
- 6 Your device has now entered the **Install Profile** page, click **More Details**, the font just added will be displayed, return to the installation page, click **Install** in the upper right corner;
- 7 Enter your device's lock screen password (if any), after confirmation, an unsigned warning message will appear, continue to click **Install** in the upper right corner, and then click the red **Install** button

again;

- ⑧ If the installation is successful, the page title will change to **Profile Installed**, click **Done**. Most applications can view and use the installed fonts.

Note

- This method cannot use WenJin Mincho as the system fallback font, that is, it cannot affect the font settings of the system interface;
- To update the font, you must first remove the old version of the font (i.e., **Remove Profile**), and then reinstall the new version of the font.

1.6 WPS Office for Android/Harmony OS

- ① Open the file manager app and find the font to be installed (fonts downloaded via browser are generally in the *Download* folder);
- ② Copy the font file to the **Internal Storage/Fonts** folder;
- ③ Open WPS Office, click **Local Fonts** in the font selection menu, and you can find the font just installed.

Note

- This method is only available in WPS Office, Xiaomi/Redmi built-in file browsing may be available;
- The specific operation method of the file manager app varies by brand, subject to actual operation. If you find that the corresponding folder cannot be found when entering the folder, you can try to enable the *Show hidden files* option;
- You can also copy the font file to the **Internal Storage/Android/data/cn.wps.moffice_eng/.cache/KingsoftOffice/.fonts** folder, but it may be cleared by the device's built-in garbage cleaning software;
- This method applies to Harmony OS version 4.3 and below;
- This method is also applicable to Xiaomi/Redmi's Hyper OS.

Chapter 2

Basic Functions

This chapter introduces the basic functions supported by WenJin Mincho. This part of the content basically does not require complicated settings. The WenJin Mincho font suite is divided into 3 parts according to planes, corresponding to Plane 0, 2, and 3 respectively.

Unicode Plane Range	
U+0000..FFFF	Basic Multilingual Plane (BMP)
U+10000..1FFFF	Supplementary Multilingual Plane (SMP)
U+20000..2FFFF	Supplementary Ideographic Plane (SIP)
U+30000..3FFFF	Tertiary Ideographic Plane (TIP)
U+E0000..EFFFF	Supplementary Special-purpose Plane (SSP)
U+F0000..FFFFF	Supplementary Private Use Area-A (SPUA-A)
U+100000..10FFFF	Supplementary Private Use Area-B (SPUA-B)

Among them, Plane 0, 2, and 3 (BMP, SIP, and TIP) contain Han ideographs.

2.1 Unicode Han Ideographs Support

2.1.1 CJK Unified Ideographs and Compatibility Ideographs Support

- Contains all Han ideographs (101,996 Unified Ideographs + 1,002 Compatibility Ideographs) defined by the current Unicode standard (version 17.0), characters defined by the Adobe-GB1 character set, and some other non-ideograph symbols. The Han ideographs and non-ideograph symbols covered by the font have met the range defined by GB 18030–2022 Implementation Level 3 (including Amendment 1). Among them, the font corresponding to Plane 0 has met the range defined by GB 18030–2022 Implementation Level 2 (including Amendment 1) [all 8,105 Han ideographs specified in the *Table of General Standard Chinese Characters* (通用规范汉字表, TGH ideographs) have been included].
- CJK Unified Ideographs, its Extension A, all of the TGH ideographs and some Han ideographs in other extension areas are presented in mainland China glyphs (aka G source), and comply with the writing of Han ideographs in the TGH ideographs, and support the pseudo G source of some Han ideographs^❶ (default mapped glyphs, no need to configure OpenType features). For the ideographs table supporting pseudo G source, please [click here to view](#).

❶ Refers to writing glyphs that do not originally have a G source in the Unicode standard according to the writing of the G source to achieve the effect of the G source.

- Glyphs that do not have a G source and do not support pseudo G source are presented in the following order of glyph standards:

G > UK > UTC > KP^❶ > H > M > T > J > K > V > SAT

Table 2.1 Unicode Han Ideographs Related Block Information

Block Name	Code Points Range	Count	Sample
CJK Unified Ideographs	4E00..9FFF	20,992	一 丁 丂 乚 𠂇 𠂉 𠂊 𠂋 𠂌 𠂍 𠂎 𠂏 𠂐 𠂑 𠂒 𠂓 𠂔 𠂕 𠂖 𠂗 𠂘 𠂙 𠂚 𠂛 𠂜 𠂝 𠂞 𠂟 𠂠 𠂡 𠂢 𠂣 𠂤 𠂥 𠂦 𠂧 𠂨 𠂩 𠂪 𠂫 𠂬 𠂭 𠂮 𠂯 𠂰 𠂱 𠂲 𠂳 𠂴 𠂵 𠂶 𠂷 𠂸 𠂹 𠂺 𠂻 𠂼 𠂽 𠂾 𠂿 𠃀 𠃁 𠃂 𠃃 𠃄 𠃅 𠃆 𠃇 𠃈 𠃉 𠃊 𠃋 𠃌 𠃍 𠃎 𠃏 𠃐 𠃑 𠃒 𠃓 𠃔 𠃕 𠃖 𠃗 𠃘 𠃙 𠃚 𠃛 𠃜 𠃝 𠃞 𠃟 𠃠 𠃡 𠃢 𠃣 𠃤 𠃥 𠃦 𠃧 𠃨 𠃩 𠃪 𠃫 𠃬 𠃭 𠃮 𠃯 𠃰 𠃱 𠃲 𠃳 𠃴 𠃵 𠃶 𠃷 𠃸 𠃹 𠃺 𠃻 𠃼 𠃽 𠃾 𠃿 𠄀 𠄁 𠄂 𠄃 𠄄 𠄅 𠄆 𠄇 𠄈 𠄉 𠄊 𠄋 𠄌 𠄍 𠄎 𠄏 𠄐 𠄑 𠄒 𠄓 𠄔 𠄕 𠄖 𠄗 𠄘 𠄙 𠄚 𠄛 𠄜 𠄝 𠄞 𠄟 𠄠 𠄡 𠄢 𠄣 𠄤 𠄥 𠄦 𠄧 𠄨 𠄩 𠄪 𠄫 𠄬 𠄭 𠄮 𠄯 𠄰 𠄱 𠄲 𠄳 𠄴 𠄵 𠄶 𠄷 𠄸 𠄹 𠄺 𠄻 𠄼 𠄽 𠄾 𠄿 𠅀 𠅁 𠅂 𠅃 𠅄 𠅅 𠅆 𠅇 𠅈 𠅉 𠅊 𠅋 𠅌 𠅍 𠅎 𠅏 𠅐 𠅑 𠅒 𠅓 𠅔 𠅕 𠅖 𠅗 𠅘 𠅙 𠅚 𠅛 𠅜 𠅝 𠅞 𠅟 𠅠 𠅡 𠅢 𠅣 𠅤 𠅥 𠅦 𠅧 𠅨 𠅩 𠅪 𠅫 𠅬 𠅭 𠅮 𠅯 𠅰 𠅱 𠅲 𠅳 𠅴 𠅵 𠅶 𠅷 𠅸 𠅹 𠅺 𠅻 𠅼 𠅽 𠅾 𠅿 𠆀 𠆁 𠆂 𠆃 𠆄 𠆅 𠆆 𠆇 𠆈 𠆉 𠆊 𠆋 𠆌 𠆍 𠆎 𠆏 𠆐 𠆑 𠆒 𠆓 𠆔 𠆕 𠆖 𠆗 𠆘 𠆙 𠆚 𠆛 𠆜 𠆝 𠆞 𠆟 𠆠 𠆡 𠆢 𠆣 𠆤 𠆥 𠆦 𠆧 𠆨 𠆩 𠆪 𠆫 𠆬 𠆭 𠆮 𠆯 𠆰 𠆱 𠆲 𠆳 𠆴 𠆵 𠆶 𠆷 𠆸 𠆹 𠆺 𠆻 𠆼 𠆽 𠆾 𠆿 𠇀 𠇁 𠇂 𠇃 𠇄 𠇅 𠇆 𠇇 𠇈 𠇉 𠇊 𠇋 𠇌 𠇍 𠇎 𠇏 𠇐 𠇑 𠇒 𠇓 𠇔 𠇕 𠇖 𠇗 𠇘 𠇙 𠇚 𠇛 𠇜 𠇝 𠇞 𠇟 𠇠 𠇡 𠇢 𠇣 𠇤 𠇥 𠇦 𠇧 𠇨 𠇩 𠇪 𠇫 𠇬 𠇭 𠇮 𠇯 𠇰 𠇱 𠇲 𠇳 𠇴 𠇵 𠇶 𠇷 𠇸 𠇹 𠇺 𠇻 𠇼 𠇽 𠇾 𠇿 𠈀 𠈁 𠈂 𠈃 𠈄 𠈅 𠈆 𠈇 𠈈 𠈉 𠈊 𠈋 𠈌 𠈍 𠈎 𠈏 𠈐 𠈑 𠈒 𠈓 𠈔 𠈕 𠈖 𠈗 𠈘 𠈙 𠈚 𠈛 𠈜 𠈝 𠈞 𠈟 𠈠 𠈡 𠈢 𠈣 𠈤 𠈥 𠈦 𠈧 𠈨 𠈩 𠈪 𠈫 𠈬 𠈭 𠈮 𠈯 𠈰 𠈱 𠈲 𠈳 𠈴 𠈵 𠈶 𠈷 𠈸 𠈹 𠈺 𠈻 𠈼 𠈽 𠈾 𠈿 𠉀 𠉁 𠉂 𠉃 𠉄 𠉅 𠉆 𠉇 𠉈 𠉉 𠉊 𠉋 𠉌 𠉍 𠉎 𠉏 𠉐 𠉑 𠉒 𠉓 𠉔 𠉕 𠉖 𠉗 𠉘 𠉙 𠉚 𠉛 𠉜 𠉝 𠉞 𠉟 𠉠 𠉡 𠉢 𠉣 𠉤 𠉥 𠉦 𠉧 𠉨 𠉩 𠉪 𠉫 𠉬 𠉭 𠉮 𠉯 𠉰 𠉱 𠉲 𠉳 𠉴 𠉵 𠉶 𠉷 𠉸 𠉹 𠉺 𠉻 𠉼 𠉽 𠉾 𠉿 𠊀 𠊁 𠊂 𠊃 𠊄 𠊅 𠊆 𠊇 𠊈 𠊉 𠊊 𠊋 𠊌 𠊍 𠊎 𠊏 𠊐 𠊑 𠊒 𠊓 𠊔 𠊕 𠊖 𠊗 𠊘 𠊙 𠊚 𠊛 𠊜 𠊝 𠊞 𠊟 𠊠 𠊡 𠊢 𠊣 𠊤 𠊥 𠊦 𠊧 𠊨 𠊩 𠊪 𠊫 𠊬 𠊭 𠊮 𠊯 𠊰 𠊱 𠊲 𠊳 𠊴 𠊵 𠊶 𠊷 𠊸 𠊹 𠊺 𠊻 𠊼 𠊽 𠊾 𠊿 𠋀 𠋁 𠋂 𠋃 𠋄 𠋅 𠋆 𠋇 𠋈 𠋉 𠋊 𠋋 𠋌 𠋍 𠋎 𠋏 𠋐 𠋑 𠋒 𠋓 𠋔 𠋕 𠋖 𠋗 𠋘 𠋙 𠋚 𠋛 𠋜 𠋝 𠋞 𠋟 𠋠 𠋡 𠋢 𠋣 𠋤 𠋥 𠋦 𠋧 𠋨 𠋩 𠋪 𠋫 𠋬 𠋭 𠋮 𠋯 𠋰 𠋱 𠋲 𠋳 𠋴 𠋵 𠋶 𠋷 𠋸 𠋹 𠋺 𠋻 𠋼 𠋽 𠋾 𠋿 𠌀 𠌁 𠌂 𠌃 𠌄 𠌅 𠌆 𠌇 𠌈 𠌉 𠌊 𠌋 𠌌 𠌍 𠌎 𠌏 𠌐 𠌑 𠌒 𠌓 𠌔 𠌕 𠌖 𠌗 𠌘 𠌙 𠌚 𠌛 𠌜 𠌝 𠌞 𠌟 𠌠 𠌡 𠌢 𠌣 𠌤 𠌥 𠌦 𠌧 𠌨 𠌩 𠌪 𠌫 𠌬 𠌭 𠌮 𠌯 𠌰 𠌱 𠌲 𠌳 𠌴 𠌵 𠌶 𠌷 𠌸 𠌹 𠌺 𠌻 𠌼 𠌽 𠌾 𠌿 𠍀 𠍁 𠍂 𠍃 𠍄 𠍅 𠍆 𠍇 𠍈 𠍉 𠍊 𠍋 𠍌 𠍍 𠍎 𠍏 𠍐 𠍑 𠍒 𠍓 𠍔 𠍕 𠍖 𠍗 𠍘 𠍙 𠍚 𠍛 𠍜 𠍝 𠍞 𠍟 𠍠 𠍡 𠍢 𠍣 𠍤 𠍥 𠍦 𠍧 𠍨 𠍩 𠍪 𠍫 𠍬 𠍭 𠍮 𠍯 𠍰 𠍱 𠍲 𠍳 𠍴 𠍵 𠍶 𠍷 𠍸 𠍹 𠍺 𠍻 𠍼 𠍽 𠍾 𠍿 𠎀 𠎁 𠎂 𠎃 𠎄 𠎅 𠎆 𠎇 𠎈 𠎉 𠎊 𠎋 𠎌 𠎍 𠎎 𠎏 𠎐 𠎑 𠎒 𠎓 𠎔 𠎕 𠎖 𠎗 𠎘 𠎙 𠎚 𠎛 𠎜 𠎝 𠎞 𠎟 𠎠 𠎡 𠎢 𠎣 𠎤 𠎥 𠎦 𠎧 𠎨 𠎩 𠎪 𠎫 𠎬 𠎭 𠎮 𠎯 𠎰 𠎱 𠎲 𠎳 𠎴 𠎵 𠎶 𠎷 𠎸 𠎹 𠎺 𠎻 𠎼 𠎽 𠎾 𠎿 𠏀 𠏁 𠏂 𠏃 𠏄 𠏅 𠏆 𠏇 𠏈 𠏉 𠏊 𠏋 𠏌 𠏍 𠏎 𠏏 𠏐 𠏑 𠏒 𠏓 𠏔 𠏕 𠏖 𠏗 𠏘 𠏙 𠏚 𠏛 𠏜 𠏝 𠏞 𠏟 𠏠 𠏡 𠏢 𠏣 𠏤 𠏥 𠏦 𠏧 𠏨 𠏩 𠏪 𠏫 𠏬 𠏭 𠏮 𠏯 𠏰 𠏱 𠏲 𠏳 𠏴 𠏵 𠏶 𠏷 𠏸 𠏹 𠏺 𠏻 𠏼 𠏽 𠏾 𠏿 𠐀 𠐁 𠐂 𠐃 𠐄 𠐅 𠐆 𠐇 𠐈 𠐉 𠐊 𠐋 𠐌 𠐍 𠐎 𠐏 𠐐 𠐑 𠐒 𠐓 𠐔 𠐕 𠐖 𠐗 𠐘 𠐙 𠐚 𠐛 𠐜 𠐝 𠐞 𠐟 𠐠 𠐡 𠐢 𠐣 𠐤 𠐥 𠐦 𠐧 𠐨 𠐩 𠐪 𠐫 𠐬 𠐭 𠐮 𠐯 𠐰 𠐱 𠐲 𠐳 𠐴 𠐵 𠐶 𠐷 𠐸 𠐹 𠐺 𠐻 𠐼 𠐽 𠐾 𠐿 𠑀 𠑁 𠑂 𠑃 𠑄 𠑅 𠑆 𠑇 𠑈 𠑉 𠑊 𠑋 𠑌 𠑍 𠑎 𠑏 𠑐 𠑑 𠑒 𠑓 𠑔 𠑕 𠑖 𠑗 𠑘 𠑙 𠑚 𠑛 𠑜 𠑝 𠑞 𠑟 𠑠 𠑡 𠑢 𠑣 𠑤 𠑥 𠑦 𠑧 𠑨 𠑩 𠑪 𠑫 𠑬 𠑭 𠑮 𠑯 𠑰 𠑱 𠑲 𠑳 𠑴 𠑵 𠑶 𠑷 𠑸 𠑹 𠑺 𠑻 𠑼 𠑽 𠑾 𠑿 𠒀 𠒁 𠒂 𠒃 𠒄 𠒅 𠒆 𠒇 𠒈 𠒉 𠒊 𠒋 𠒌 𠒍 𠒎 𠒏 𠒐 𠒑 𠒒 𠒓 𠒔 𠒕 𠒖 𠒗 𠒘 𠒙 𠒚 𠒛 𠒜 𠒝 𠒞 𠒟 𠒠 𠒡 𠒢 𠒣 𠒤 𠒥 𠒦 𠒧 𠒨 𠒩 𠒪 𠒫 𠒬 𠒭 𠒮 𠒯 𠒰 𠒱 𠒲 𠒳 𠒴 𠒵 𠒶 𠒷 𠒸 𠒹 𠒺 𠒻 𠒼 𠒽 𠒾 𠒿 𠓀 𠓁 𠓂 𠓃 𠓄 𠓅 𠓆 𠓇 𠓈 𠓉 𠓊 𠓋 𠓌 𠓍 𠓎 𠓏 𠓐 𠓑 𠓒 𠓓 𠓔 𠓕 𠓖 𠓗 𠓘 𠓙 𠓚 𠓛 𠓜 𠓝 𠓞 𠓟 𠓠 𠓡 𠓢 𠓣 𠓤 𠓥 𠓦 𠓧 𠓨 𠓩 𠓪 𠓫 𠓬 𠓭 𠓮 𠓯 𠓰 𠓱 𠓲 𠓳 𠓴 𠓵 𠓶 𠓷 𠓸 𠓹 𠓺 𠓻 𠓼 𠓽 𠓾 𠓿 𠔀 𠔁 𠔂 𠔃 𠔄 𠔅 𠔆 𠔇 𠔈 𠔉 𠔊 𠔋 𠔌 𠔍 𠔎 𠔏 𠔐 𠔑 𠔒 𠔓 𠔔 𠔕 𠔖 𠔗 𠔘 𠔙 𠔚 𠔛 𠔜 𠔝 𠔞 𠔟 𠔠 𠔡 𠔢 𠔣 𠔤 𠔥 𠔦 𠔧 𠔨 𠔩 𠔪 𠔫 𠔬 𠔭 𠔮 𠔯 𠔰 𠔱 𠔲 𠔳 𠔴 𠔵 𠔶 𠔷 𠔸 𠔹 𠔺 𠔻 𠔼 𠔽 𠔾 𠔿 𠕀 𠕁 𠕂 𠕃 𠕄 𠕅 𠕆 𠕇 𠕈 𠕉 𠕊 𠕋 𠕌 𠕍 𠕎 𠕏 𠕐 𠕑 𠕒 𠕓 𠕔 𠕕 𠕖 𠕗 𠕘 𠕙 𠕚 𠕛 𠕜 𠕝 𠕞 𠕟 𠕠 𠕡 𠕢 𠕣 𠕤 𠕥 𠕦 𠕧 𠕨 𠕩 𠕪 𠕫 𠕬 𠕭 𠕮 𠕯 𠕰 𠕱 𠕲 𠕳 𠕴 𠕵 𠕶 𠕷 𠕸 𠕹 𠕺 𠕻 𠕼 𠕽 𠕾 𠕿 𠖀 𠖁 𠖂 𠖃 𠖄 𠖅 𠖆 𠖇 𠖈 𠖉 𠖊 𠖋 𠖌 𠖍 𠖎 𠖏 𠖐 𠖑 𠖒 𠖓 𠖔 𠖕 𠖖 𠖗 𠖘 𠖙 𠖚 𠖛 𠖜 𠖝 𠖞 𠖟 𠖠 𠖡 𠖢 𠖣 𠖤 𠖥 𠖦 𠖧 𠖨 𠖩 𠖪 𠖫 𠖬 𠖭 𠖮 𠖯 𠖰 𠖱 𠖲 𠖳 𠖴 𠖵 𠖶 𠖷 𠖸 𠖹 𠖺 𠖻 𠖼 𠖽 𠖾 𠖿 𠗀 𠗁 𠗂 𠗃 𠗄 𠗅 𠗆 𠗇 𠗈 𠗉 𠗊 𠗋 𠗌 𠗍 𠗎 𠗏 𠗐 𠗑 𠗒 𠗓 𠗔 𠗕 𠗖 𠗗 𠗘 𠗙 𠗚 𠗛 𠗜 𠗝 𠗞 𠗟 𠗠 𠗡 𠗢 𠗣 𠗤 𠗥 𠗦 𠗧 𠗨 𠗩 𠗪 𠗫 𠗬 𠗭 𠗮 𠗯 𠗰 𠗱 𠗲 𠗳 𠗴 𠗵 𠗶 𠗷 𠗸 𠗹 𠗺 𠗻 𠗼 𠗽 𠗾 𠗿 𠘀 𠘁 𠘂 𠘃 𠘄 𠘅 𠘆 𠘇 𠘈 𠘉 𠘊 𠘋 𠘌 𠘍 𠘎 𠘏 𠘐 𠘑 𠘒 𠘓 𠘔 𠘕 𠘖 𠘗 𠘘 𠘙 𠘚 𠘛 𠘜 𠘝 𠘞 𠘟 𠘠 𠘡 𠘢 𠘣 𠘤 𠘥 𠘦 𠘧 𠘨 𠘩 𠘪 𠘫 𠘬 𠘭 𠘮 𠘯 𠘰 𠘱 𠘲 𠘳 𠘴 𠘵 𠘶 𠘷 𠘸 𠘹 𠘺 𠘻 𠘼 𠘽 𠘾 𠘿 𠙀 𠙁 𠙂 𠙃 𠙄 𠙅 𠙆 𠙇 𠙈 𠙉 𠙊 𠙋 𠙌 𠙍 𠙎 𠙏 𠙐 𠙑 𠙒 𠙓 𠙔 𠙕 𠙖 𠙗 𠙘 𠙙 𠙚 𠙛 𠙜 𠙝 𠙞 𠙟 𠙠 𠙡 𠙢 𠙣 𠙤 𠙥 𠙦 𠙧 𠙨 𠙩 𠙪 𠙫 𠙬 𠙭 𠙮 𠙯 𠙰 𠙱 𠙲 𠙳 𠙴 𠙵 𠙶 𠙷 𠙸 𠙹 𠙺 𠙻 𠙼 𠙽 𠙾 𠙿 𠚀 𠚁 𠚂 𠚃 𠚄 𠚅 𠚆 𠚇 𠚈 𠚉 𠚊 𠚋 𠚌 𠚍 𠚎 𠚏 𠚐 𠚑 𠚒 𠚓 𠚔 𠚕 𠚖 𠚗 𠚘 𠚙 𠚚 𠚛 𠚜 𠚝 𠚞 𠚟 𠚠 𠚡 𠚢 𠚣 𠚤 𠚥 𠚦 𠚧 𠚨 𠚩 𠚪 𠚫 𠚬 𠚭 𠚮 𠚯 𠚰 𠚱 𠚲 𠚳 𠚴 𠚵 𠚶 𠚷 𠚸 𠚹 𠚺 𠚻 𠚼 𠚽 𠚾 𠚿 𠛀 𠛁 𠛂 𠛃 𠛄 𠛅 𠛆 𠛇 𠛈 𠛉 𠛊 𠛋 𠛌 𠛍 𠛎 𠛏 𠛐 𠛑 𠛒 𠛓 𠛔 𠛕 𠛖 𠛗 𠛘 𠛙 𠛚 𠛛 𠛜 𠛝 𠛞 𠛟 𠛠 𠛡 𠛢 𠛣 𠛤 𠛥 𠛦 𠛧 𠛨 𠛩 𠛪 𠛫 𠛬 𠛭 𠛮 𠛯 𠛰 𠛱 𠛲 𠛳 𠛴 𠛵 𠛶 𠛷 𠛸 𠛹 𠛺 𠛻 𠛼 𠛽 𠛾 𠛿 𠜀 𠜁 𠜂 𠜃 𠜄 𠜅 𠜆 𠜇 𠜈 𠜉 𠜊 𠜋 𠜌 𠜍 𠜎 𠜏 𠜐 𠜑 𠜒 𠜓 𠜔 𠜕 𠜖 𠜗 𠜘 𠜙 𠜚 𠜛 𠜜 𠜝 𠜞 𠜟 𠜠 𠜡 𠜢 𠜣 𠜤 𠜥 𠜦 𠜧 𠜨 𠜩 𠜪 𠜫 𠜬 𠜭 𠜮 𠜯 𠜰 𠜱 𠜲 𠜳 𠜴 𠜵 𠜶 𠜷 𠜸 𠜹 𠜺 𠜻 𠜼 𠜽 𠜾 𠜿 𠝀 𠝁 𠝂 𠝃 𠝄 𠝅 𠝆 𠝇 𠝈 𠝉 𠝊 𠝋 𠝌 𠝍 𠝎 𠝏 𠝐 𠝑 𠝒 𠝓 𠝔 𠝕 𠝖 𠝗 𠝘 𠝙 𠝚 𠝛 𠝜 𠝝 𠝞 𠝟 𠝠 𠝡 𠝢 𠝣 𠝤 𠝥 𠝦 𠝧 𠝨 𠝩 𠝪 𠝫 𠝬 𠝭 𠝮 𠝯 𠝰 𠝱 𠝲 𠝳 𠝴 𠝵 𠝶 𠝷 𠝸 𠝹 𠝺 𠝻 𠝼 𠝽 𠝾 𠝿 𠞀 𠞁 𠞂 𠞃 𠞄 𠞅 𠞆 𠞇 𠞈 𠞉 𠞊 𠞋 𠞌 𠞍 𠞎 𠞏 𠞐 𠞑 𠞒 𠞓 𠞔 𠞕 𠞖 𠞗 𠞘 𠞙 𠞚 𠞛 𠞜 𠞝 𠞞 𠞟 𠞠 𠞡 𠞢 𠞣 𠞤 𠞥 𠞦 𠞧 𠞨 𠞩 𠞪 𠞫 𠞬 𠞭 𠞮 𠞯 𠞰 𠞱 𠞲 𠞳 𠞴 𠞵 𠞶 𠞷 𠞸 𠞹 𠞺 𠞻 𠞼 𠞽 𠞾 𠞿 𠟀 𠟁 𠟂 𠟃 𠟄 𠟅 𠟆 𠟇 𠟈 𠟉 𠟊 𠟋 𠟌 𠟍 𠟎 𠟏 𠟐 𠟑 𠟒 𠟓 𠟔 𠟕 𠟖 𠟗 𠟘 𠟙 𠟚 𠟛 𠟜 𠟝 𠟞 𠟟 𠟠 𠟡 𠟢 𠟣 𠟤 𠟥 𠟦 𠟧 𠟨 𠟩 𠟪 𠟫 𠟬 𠟭 𠟮 𠟯 𠟰 𠟱 𠟲 𠟳 𠟴 𠟵 𠟶 𠟷 𠟸 𠟹 𠟺 𠟻 𠟼 𠟽 𠟾 𠟿 𠠀 𠠁 𠠂 𠠃 𠠄 𠠅 𠠆 𠠇 𠠈 𠠉 𠠊 𠠋 𠠌 𠠍 𠠎 𠠏 𠠐 𠠑 𠠒 𠠓 𠠔 𠠕 𠠖 𠠗 𠠘 𠠙 𠠚 𠠛 𠠜 𠠝 𠠞 𠠟 𠠠 𠠡 𠠢 𠠣 𠠤 𠠥 𠠦 𠠧 𠠨 𠠩 𠠪 𠠫 𠠬 𠠭 𠠮 𠠯 𠠰 𠠱 𠠲 𠠳 𠠴 𠠵 𠠶 𠠷 𠠸 𠠹 𠠺 𠠻 𠠼 𠠽 𠠾 𠠿 𠡀 𠡁 𠡂 𠡃 𠡄 𠡅 𠡆 𠡇 𠡈 𠡉 𠡊 𠡋 𠡌 𠡍 𠡎 𠡏 𠡐 𠡑 𠡒 𠡓 𠡔 𠡕 𠡖 𠡗 𠡘 𠡙 𠡚 𠡛 𠡜 𠡝 𠡞 𠡟 𠡠 𠡡 𠡢 𠡣 𠡤 𠡥 𠡦 𠡧 𠡨 𠡩 𠡪 𠡫 𠡬 𠡭 𠡮 𠡯 𠡰 𠡱 𠡲 𠡳 𠡴 𠡵 𠡶 𠡷 𠡸 𠡹 𠡺 𠡻 𠡼 𠡽 𠡾 𠡿 𠢀 𠢁 𠢂 𠢃 𠢄 𠢅 𠢆 𠢇 𠢈 𠢉 𠢊 𠢋 𠢌 𠢍 𠢎 𠢏 𠢐 𠢑 𠢒 𠢓 𠢔 𠢕 𠢖 𠢗 𠢘 𠢙 𠢚 𠢛 𠢜 𠢝 𠢞 𠢟 𠢠 𠢡 𠢢 𠢣 𠢤 𠢥 𠢦 𠢧 𠢨 𠢩 𠢪 𠢫 𠢬 𠢭 𠢮 𠢯 𠢰 𠢱 𠢲 𠢳 𠢴 𠢵 𠢶 𠢷 𠢸 𠢹 𠢺 𠢻 𠢼 𠢽 𠢾 𠢿 𠣀 𠣁 𠣂 𠣃 𠣄 𠣅 𠣆 𠣇 𠣈 𠣉 𠣊 𠣋 𠣌 𠣍 𠣎 𠣏 𠣐 𠣑 𠣒 𠣓 𠣔 𠣕 𠣖 𠣗 𠣘 𠣙 𠣚 𠣛 𠣜 𠣝 𠣞 𠣟 𠣠 𠣡 𠣢 𠣣 𠣤 𠣥 𠣦 𠣧 𠣨 𠣩 𠣪 𠣫 𠣬 𠣭 𠣮 𠣯 𠣰 𠣱 𠣲 𠣳 𠣴 𠣵 𠣶 𠣷 𠣸 𠣹 𠣺 𠣻 𠣼 𠣽 𠣾 𠣿 𠤀 𠤁 𠤂 𠤃 𠤄 𠤅 𠤆 𠤇 𠤈 𠤉 𠤊 𠤋 𠤌 𠤍 𠤎 𠤏 𠤐 𠤑 𠤒 𠤓 𠤔 𠤕 𠤖 𠤗 𠤘 𠤙 𠤚 𠤛 𠤜 𠤝 𠤞 𠤟 𠤠 𠤡 𠤢 𠤣 𠤤 𠤥 𠤦 𠤧 𠤨 𠤩 𠤪 𠤫 𠤬 𠤭 𠤮 𠤯 𠤰 𠤱 𠤲 𠤳 𠤴 𠤵 𠤶 𠤷 𠤸 𠤹 𠤺 𠤻 𠤼 𠤽 𠤾 𠤿 𠥀 𠥁 𠥂 𠥃 𠥄 𠥅 𠥆 𠥇 𠥈 𠥉 𠥊 𠥋 𠥌 𠥍 𠥎 𠥏 𠥐 𠥑 𠥒 𠥓 𠥔 𠥕 𠥖 𠥗 𠥘 𠥙 𠥚 𠥛 𠥜 𠥝 𠥞 𠥟 𠥠 𠥡 𠥢 𠥣 𠥤 𠥥 𠥦 𠥧 𠥨 𠥩 𠥪 𠥫 𠥬 𠥭 𠥮 𠥯 𠥰 𠥱 𠥲 𠥳 𠥴 𠥵 𠥶 𠥷 𠥸 𠥹 𠥺 𠥻 𠥼 𠥽 𠥾 𠥿 𠦀 𠦁 𠦂 𠦃 𠦄 𠦅 𠦆 𠦇 𠦈 𠦉 𠦊 𠦋 𠦌 𠦍 𠦎 𠦏 𠦐 𠦑 𠦒 𠦓 𠦔 𠦕 𠦖 𠦗 𠦘 𠦙 𠦚 𠦛 𠦜 𠦝 𠦞 𠦟 𠦠 𠦡 𠦢 𠦣 𠦤 𠦥 𠦦 𠦧 𠦨 𠦩 𠦪 𠦫 𠦬 𠦭 𠦮 𠦯 𠦰 𠦱 𠦲 𠦳 𠦴 𠦵 𠦶 𠦷 𠦸 𠦹 𠦺 𠦻 𠦼 𠦽 𠦾 𠦿 𠧀 𠧁 𠧂 𠧃 𠧄 𠧅 𠧆 𠧇 𠧈 𠧉 𠧊 𠧋 𠧌 𠧍 𠧎 𠧏 𠧐 𠧑 𠧒 𠧓 𠧔 𠧕 𠧖 𠧗 𠧘 𠧙 𠧚 𠧛 𠧜 𠧝 𠧞 𠧟 𠧠 𠧡 𠧢 𠧣 𠧤 𠧥 𠧦 𠧧 𠧨 𠧩 𠧪 𠧫 𠧬 𠧭 𠧮 𠧯 𠧰 𠧱

Table 2.2 Unicode IVD Source Information

Source Name	Count	Sample
Adobe-Japan1 (Adobe-Japan1 Character Collection)	14,684	𪛗𪛗𪛗メ……罐衰蹊示
Hanyo-Denshi (汎用電子)	13,045	𪛗𪛗𪛗𪛗……重重鼈鼈
Moji_Joho (Moji Joho Kiban Project, 文字情報基盤整備事業)	11,384	𪛗𪛗𪛗𪛗……庾庶釐釐
KRName (Korean personal name)	36	卿卿屋屋……鉸鉸龜龜
MSARG (Macao SAR Government)	154	姘姘姘踎……襠襠倂倂
CAAPH (Culture and Art Publishing House)	198	婁婁科科……滄歌歌歌

- Supports character display and glyph combination of combining symbols for romanization systems (Latin alphabet) such as Hànyǔ Pīnyīn, Mǐnnán Pēh-ōe-jī (Minnan POJ), Hakká Phák-fa-sṳ̌(Hakka PFS), Mǐnnán Hǎikou Pēh-ōe-jī (Minnan Haikou POJ), Matsu Mindong Romanization, etc.
- Supports Zhuyin symbol display for Mandarin (including Old National Pronunciation, New National Pronunciation), Minnan, Hakka, and Cantonese. Some Zhuyin symbols not included in Unicode can be invoked through OpenType features (see Section 3.2.4, page 16).
- Reference for romanization and Zhuyin systems supported by this font: [IMing Functional Report Version 8.100 \(in Chinese\)](#).

[illegible]

❶ The sample text is taken from the “Earth” entry in Wikipedia of the corresponding language. The content converted into other forms (such as IPA, romanization, and Zhuyin forms) mentioned later is converted from this source and will not be repeated.

Introduction to OpenType Features

 **Tip**
Not all software can invoke the OpenType features of the font and achieve good display results. Older systems and application software may not even support OpenType features.

3.1.1 Standard Glyph Switching

By invoking the *Character Variant/cvXX* OpenType feature, standard glyph switching for Han ideographs can be achieved.

Since a few IVD glyphs share a glyph with the replacement glyph of this feature, if this feature is used together with IVD variants, it may affect the IVD glyphs. Therefore, it is recommended that users disable this feature when displaying IVD glyphs.

- Default 𪚩𪚪𪚫𪚬𪚭𪚮𪚯𪚰𪚱𪚲𪚳𪚴𪚵𪚶𪚷𪚸𪚹𪚺𪚻𪚼𪚽𪚾𪚿𪚩𪚪𪚫𪚬𪚭𪚮𪚯𪚰𪚱𪚲𪚳𪚴𪚵𪚶𪚷𪚸𪚹𪚺𪚻𪚼𪚽𪚾𪚿
- cv01 𪚩𪚪𪚫𪚬𪚭𪚮𪚯𪚰𪚱𪚲𪚳𪚴𪚵𪚶𪚷𪚸𪚹𪚺𪚻𪚼𪚽𪚾𪚿𪚩𪚪𪚫𪚬𪚭𪚮𪚯𪚰𪚱𪚲𪚳𪚴𪚵𪚶𪚷𪚸𪚹𪚺𪚻𪚼𪚽𪚾𪚿

3.1.3 Vietnamese Tone Mark Combination

This part is supported by all sub-plane (Plane 0, 2, 3) fonts. Generally, no special setting is required, and the software enables it automatically.

By appending “◌ ˆ” (U+16FF0) or “◌ ˆ” (U+16FF1) after some Han ideographs, the combined Han ideographs can be generated.

This feature relies on the *Glyph Composition / Decomposition/ccmp* OpenType feature tag. Generally, the software will automatically enable this feature.

午 (U+5348) + ◌ ˆ (U+16FF0) → 𠂔(U+2A7F3)	𠂔 (U+5BFE) + ◌ ˆ (U+16FF0) → 𠂔(U+2BD2C)
呂 (U+5415) + ◌ ˆ (U+16FF0) → 𠂔(U+2A849)	魚 (U+9B5A) + ◌ ˆ (U+16FF0) → 𩺰(U+29D5C)
呂 (U+5442) + ◌ ˆ (U+16FF0) → 𠂔(U+2A849)	𠂔 (U+3193D) + ◌ ˆ (U+16FF0) → 𠂔(U+2A809)

3.2 OpenType Features for Romanization, Zhuyin, IPA Symbols

3.2.1 Hànyǔ Pīnyīn Handwriting Support

By invoking the *Stylistic Set 1/ss01* OpenType feature, the handwriting style conversion of Hànyǔ Pīnyīn letters *a* and *g* can be achieved, and the form of the rising tone mark “◌ ˆ” is also replaced accordingly (stroke direction: ↗⇒↘).

Default	Dìqiú shì mùqián tàiyángxì zhōng yǐ tàiyáng wéi zhōngxīn yóunèi-xiàngwài de dìsānkē xíngxīng.
ss01	Dìqiú shì mùqián tàiyángxì zhōng yǐ tàiyáng wéi zhōngxīn yóunèi-xiàngwài de dìsānkē xíngxīng.
Red characters have changes compared to the default glyphs (when relevant features are not enabled).	

3.2.2 Superscript and Subscript Numbers

By invoking the *Superscript/sups* or *Subscript/subs* OpenType feature, superscript or subscript display of numbers can be achieved.

Default	0123456789	sups	0123456789
		subs	0123456789

3.2.3 Free Switching of Zhuyin Tone Mark Forms

By invoking the *Stylistic Set 2, 3, 4/ss02, ss03, ss04* OpenType features, the style of Zhuyin tone marks can be changed.

- **Default** The Zhuyin tone mark itself occupies the glyph width, and the tone mark is placed in a higher position;

- **ss02** The Zhuyin tone mark is displayed in full width, occupying the width of a whole Han ideograph, and the tone mark is placed in the center;
- **ss03** The Zhuyin tone mark is placed in the upper right corner of the Zhuyin symbol;
- **ss04** The Zhuyin tone mark is placed directly above the Zhuyin symbol and does not occupy the glyph width (i.e., the glyph width is zero). At the same time, the Zhuyin symbol “一” is displayed as a vertical line (i.e., the form used in dictionaries in mainland China);

[illegible]

3.2.4 Support for Some Zhuyin Symbols Not Included in Unicode

By invoking the *Character Variant 11*/cv11 and *Subscript*/subs OpenType features, the input of some Zhuyin symbols not included in Unicode can be achieved.

Under the **Character Variant 11/cv11** or **Subscript/subs** OpenType feature, the subscript Zhuyin symbols “ㄣㄣㄣㄣㄣ” will be placed on the upper right side of the Han ideograph frame (i.e., the lower right side of the previous rhyme ending) in a vertical layout environment, rather than the right side of the previous rhyme ending.

Table 3.1 Some Zhuyin Symbols Not Included in Unicode

Unicode	3117	3118	3119	311A	312A	31B0	31B1	31B2	31B4	31B5	31BE
Default	𐤑	𐤒	𐤓	𐤔	𐤕	𐤖	𐤗	𐤘	𐤙	𐤚	𐤛
cv11	𐤑	𐤒	𐤓	𐤔	𐤕	𐤖	𐤗	𐤘	𐤙	𐤚	𐤛
Unicode	3105	3106	3107	3109	310A	310B	310D	310E	310F	312B	
Default	𐤜	𐤝	𐤞	𐤟	𐤠	𐤡	𐤢	𐤣	𐤤	𐤥	
subs	𐤜	𐤝	𐤞	𐤟	𐤠	𐤡	𐤢	𐤣	𐤤	𐤥	

Red characters have been included in Unicode.

3.2.8 Form Conversion of Four-Corner Tone Marks

By invoking the *Character Variant 12/cv12* OpenType feature, the four-corner tone marks for Han ideographs can be converted from circles to dots.

Default	平上去入 永
cv12	平上去入 永

3.3 Non-CJK Language Related OpenType Features

3.3.1 Non-CJK Punctuation Display

When the language is set to non-CJK (or automatically identified as a non-CJK text segment), some punctuation marks will automatically adjust to non-CJK styles (proportional width).

This feature relies on the *Localized Forms/locl* OpenType feature tag. Generally, the software will automatically enable this feature. As long as the language of the text segment is set correctly, it can be automatically replaced with the correct form for the corresponding language. However, some older software may not support it.

Single and double quotes (“”) can also be followed by VS characters VS1, VS2 (U+FE00, FE01) to achieve style conversion, which is not affected by OpenType features.

 **Warning**

The Windows GDI compatible version (**ended support**) does not have this feature. To avoid the BUG where punctuation marks are displayed as non-CJK punctuation in all cases in the Windows GDI environment, users using the Windows GDI compatible version need to manually enable the OpenType *Proportional Widths/pwid* feature to make the punctuation marks change form. The conversion of single and double quotes (“”) via VS is not affected.

Default (CJK)	Kinich ^❶ sighed. “Ajaw, be quiet.” “Fools!” K’ uhul·Ajaw ^❷ mocked… “Read § 1-9 of my rules—his power is ≤0‰ and mine is ≥999! My intellect ×10÷2−1≈infinity±0, proving we are ≠. You ‘servants’ are pathetic!” [Audio·Offline^]
Non-CJK Env	Kinich sighed. “Ajaw, be quiet.” “Fools!” K’uhul·Ajaw mocked... “Read §1-9 of my rules—his power is ≤0‰ and mine is ≥999! My intellect ×10÷2−1≈infinity±0, proving we are ≠. You ‘servants’ are pathetic!” [Audio·Offline^]

❶ Kinich (Playable character in *Genshin Impact*): <https://genshin-impact.fandom.com/wiki/Kinich>
❷ K’uhul Ajaw (Character in *Genshin Impact*): <https://genshin-impact.fandom.com/wiki/Ajaw>

Blue symbols have changes compared to the default glyphs (when relevant features are not enabled).

 Warning

In the HarfBuzz environment, pure symbol fragments (excluding letters and Han ideographs, such as “≥§≤”) may fail to convert the default full-width symbols to proportional-width forms (the VS feature for single and double quotes remains unaffected). Users can manually specify the script of the text segment (if possible) or manually enable the OpenType *Proportional Widths*/pwid feature to avoid this issue.

Technical Explanation: Before processing a text fragment, HarfBuzz determines the default script of that fragment based on the Unicode properties of the characters (such as **Letter**, **Common**, etc.). If a fragment contains letters or Han ideographs, HarfBuzz will set its default script to the corresponding script (like **latn**, **hani**, etc.), allowing the correct application of OpenType features for substitution. However, if a fragment contains only symbol characters (and no script is manually specified), HarfBuzz will set its script to **DFLT** (default script), which means it will only apply OpenType features that are active by default and will not apply script-specific OpenType features (like **latn**, **hani**, etc.).

3.3.2 Language-Dependent Contextual Substitution

This font will automatically adapt to the appropriate contextual automatic substitution features based on the current language environment. This feature relies on the *Localized Forms*/locl OpenType feature tag.

 Warning

The Windows GDI compatible version (**ended support**) may need to manually enable the OpenType *Proportional Widths*/pwid feature to present the best effect.

Default (English)	ÍJ íj COL·LEGI paral·lel fill Şş
Dutch	ÍJ íj COL·LEGI paral·lel fill Şş
Catalan	ÍJ íj COL·LEGI para H el fill Şş
Turkish	ÍJ íj COL·LEGI paral·lel f ill Şş
Romanian	ÍJ íj COL·LEGI paral·lel fill Şş
Default (Russian, Ukrainian)	ИЙЛФбвгджзийклнптфцчщъью
Bulgarian	ИЙЛФбвгджзийклнптфцчщъью
Serbian	ИЙЛФ б вгджзийклнптфцчщъью

Glyphs marked with other colors have changes compared to the default glyphs (when relevant features are not enabled).

3.3.3 Italic Character Substitution

By invoking the ***Italics***/***ital*** OpenType feature, the italic form of Latin letters can be displayed. Most other OpenType features are still supported in the italic form.

The italic form currently only supports the GF Latin Core character set and characters in the various romanization systems supported above (Vietnamese is not yet supported).

Hànyǔ Pīnyīn	<i>Dìqiú shì mùqián tàiyángxì zhōng yǐ tàiyáng wéi zhōngxīn yóunèixiàngwài de dìsānkē xíngxīng.</i>
Cantonese Jyutping	<i>Dei⁶kau⁴ hai⁶ taai³yeung⁴hai⁶ yap⁶min⁶ ge³ yat¹lap¹ haang⁴sing¹.</i>
Minnan POJ	<i>Tē-kiū sī Thài-iông-hē 9 tōa hêng-chheⁿ chi-it.</i>
Hakka PFS	<i>Thi-khiù he Thai-yòng-hi chûng-hàng sên chû-yit.</i>
Minnan Haikou POJ	<i>ÓÒÒÒÕÕÕÕ ớồồồốốốố ỠỠỠỠ ỡữữữ</i>
Matsu Mindong Romanization	<i>Dê-giù sê tái-iòng-hiê gì dậ sǎng lǎk giàng-sǎng.</i>
English	<i>The quick brown fox jumps over the lazy dog.</i>
French	<i>Portez ce vieux whisky au juge blond qui fume.</i>
Spanish	<i>El veloz murciélago hindú comía feliz cardillo y kiwi. La cigüeña tocaba el saxofón detrás del palenque de paja.</i>
Czech	<i>Přiliš žlutoučký kůň úpěl ďábelské ódy.</i>
Icelandic	<i>Kæmi ný öxi hér, ykist þjófum nú bæði víl og ádrepa.</i>
Polish	<i>Pchnąc w tę łódź jeża lub ośm skrzyń fig.</i>
Romanian	<i>Încă vând gem, whisky bej și tequila roz, preț fix.</i>
Turkish	<i>Pijamalı hasta yağız şoföre çabucak güvendi.</i>
Hànyǔ Pīnyīn Handwriting Support	<i>Dìqiú^ú shì mùqián^{án} tàiyáng^{ng}xì zhōng^g yǐ tàiyáng^{ng} wéi zhōng^{gx}xīn yóunèixiàng^{gw}wài de dìsānkē^ē xíng^{gx}xīng^g.</i>
Language-Dependent Contextual Substitution	<i>Íí íj COLLEGI parallel fill Şş</i>

3.3.4 Slashed Zero

By invoking the ***Slashed Zero***/***zero*** OpenType feature, the number 0 can be displayed with a slash.

In addition, you can also append VS1 (U+FE00) after the number 0 to invoke the slashed form.

Table 3.3 List of Slashed Zero Styles

Unicode Code Point	0030	0030 (+ital)	FF10
Default	0	0	0
zero	0	0	0

3.3.5 Fraction Related OpenType Features

By invoking the ***Fractions***/frac OpenType feature, automatic conversion of slashed fractions (such as 1/2) can be achieved.

At the same time, ***Numerators***/numr and ***Denominators***/dnom features are also provided (different from superscript/subscript numbers), allowing users to determine the characteristics of fractions as needed.

This feature does not support the conversion of fractions in italic numbers.

Default	0123456789 114.514/1,919,810	numr	0123456789 114.514/1,919,810
frac	0123456789 114.514/1,919,810	dnom	0123456789 114.514/1,919,810

3.3.6 Ordinal Form Substitution

By invoking the ***Ordinals***/ordn OpenType feature, the ordinal forms of letters *a* and *o* can be substituted.

In Spanish, if a number is followed by a superscript *a* or *o*, it indicates that the word is an ordinal number.

Default	12.a 12.o 13.a 13.o
ordn	12. ^a 12. ^o 13. ^a 13. ^o

Red symbols have changes compared to the default glyphs (when relevant features are not enabled).

3.3.7 Tabular Figures Support

By invoking the ***Tabular Figures***/tnum OpenType feature, the tabular style of numbers can be substituted. Full-width numbers are not affected, and the width of each full-width number is always one Chinese character width.

Default	01234567890	01234567890	0 1 2 3 4 5 6 7 8 9 0
tnum	01234567890	01234567890	0 1 2 3 4 5 6 7 8 9 0

3.4 Other OpenType Features Introduction

3.4.1 Punctuation Kerning Support

By invoking the ***Contextual Half-width Spacing***/chws OpenType feature, kerning of horizontal punctuation marks can be achieved. In a vertical layout environment, invoking the ***Vertical Contextual Half-width Spacing***/vchw OpenType feature can achieve kerning of vertical punctuation marks. This feature is automatically enabled by default in Chrome/Chromium/Edge 123.0 and above browsers.

In addition, WenJin Mincho also configures ***Alternate Half Widths***/halt and ***Alternate Vertical Half Metrics***/vhal features for punctuation marks, allowing punctuation marks to be presented in half-width forms.

 Warning

In a vertical layout environment, due to the 90° rotation characteristic of non-CJK characters, some punctuation marks may not be kerned correctly.

	Default	chws	halt
Default	‘ ° ? ! (□ ‘ ”	‘ ° ? ! (□ ‘ ”	‘ ° ? ! (□ ‘ ”
chws	‘ ° ? ! (□ ‘ ”	‘ ° ? ! (□ ‘ ”	‘ ° ? ! (□ ‘ ”
halt	‘ ° ? ! (□ ‘ ”	‘ ° ? ! (□ ‘ ”	‘ ° ? ! (□ ‘ ”

3.4.2 Centered Punctuation Display

By invoking the ***Centered CJK Punctuation***/cpct OpenType feature, punctuation marks can be displayed in the center.

In addition, you can also append VS2 (U+FE01) after the punctuation mark to achieve the above effect.

Default	、 。 ! , . : ; ?
cpct	、 。 ! ’ . : ; ?

3.4.3 Vertical Punctuation Display

By invoking the *Stylistic Set 20/ss20* OpenType feature, punctuation marks can be placed on the right side of Chinese characters in a vertical layout environment.

Default	清明 [唐] 杜牧
	清明时节雨纷纷， 路上行人欲断魂。 借问酒家何处有， 牧童遥指杏花村。
ss20	清明 [唐] 杜牧
	清明时节雨纷纷， 路上行人欲断魂。 借问酒家何处有， 牧童遥指杏花村。

3.4.4 Full-width/Half-width/Proportional Character Switching

By invoking the **Full Widths**/fwid, **Half Widths**/hwid, and **Proportional Widths**/pwid OpenType features, free switching of full-width/half-width/proportional forms of ASCII characters, Hànyǔ Pīnyīn letters, and a few symbols can be achieved.

Default	!"*0123ABCDabcdāóêîñü¥€ ¤ % & ' 4 E f £
fwid	! " + * 0 1 2 3 A B C D a b c d ā ó ê î ñ ü ¥ € ¤ % & ' 4 E f £
hwid	!"*0123ABCDabcdāóêîñü¥€ ¤ % & ' 4 E f £
pwid	!"*0123ABCDabcdāóêîñü¥€ ¤ % & ' 4 E f £

Red, Green, and Blue characters have changes compared to the default glyphs (when relevant features are not enabled).

3.4.5 Discretionary Ligatures

By invoking the *Discretionary Ligatures*/dlig OpenType feature, automatic combination of some ligatures can be achieved.

Default	mg °C c/o 12月 31日 24点
dlig	mg °C % 12月 31日 24点

3.4.6 Squared Numbers and Letters

By invoking the *Stylistic Set 10*/ss10 OpenType feature, the conversion from circled numbers/letters to squared numbers/letters can be achieved.

Default	①②③④…⑤⑩ a b c … z
ss10	0 1 2 3 4 … 50 a b c … z

3.4.7 Alternate Annotation Forms for Ordinal Characters

By invoking the *Alternate Annotation Forms*/nalt OpenType feature, the conversion of ordinal characters (numbers, letters, ordinal Han ideographs) to their alternate annotation forms can be achieved.

Default	0123456 ABCDEFG abcdefg 一 二 三 四 五 六 七 八 九 十
nalt	0 1 2 3 4 5 6 A B C D E F G a b c d e f g 一 二 三 四 五 六 七 八 九 十 ①②③④⑤⑥ A(B)(C)(D)(E)(F)(G) a(b)(c)(d)(e)(f)(g) 一(二)(三)(四)(五)(六)(七)(八)(九)(十) 0 1 2 3 4 5 6 A B C D E F G a b c d e f g 一(二)(三)(四)(五)(六)(七)(八)(九)(十) ①(1)(2)(3)(4)(5)(6) A(B)(C)(D)(E)(F)(G) a(b)(c)(d)(e)(f)(g) 一 二 三 四 五 六 七 八 九 十 01.2.3.4.5.6. ABCDEFG abcdefg 一 二 三 四 五 六 七 八 九 十 01. 2. 3. 4. 5. 6. ABCDEFG abcdefg 一 二 三 四 五 六 七 八 九 十

Chapter 4

OpenType Feature Usage Guide

⚠ Warning

If you are using the Condensed Line-Spacing Version (C Version)/Windows GDI Compatible Version (W Version) of the font, you should modify the following code content as needed. If you need to call it locally in a web page or develop a $\LaTeX$ macro package, you can consider the Standard Version, C Version, and W Version simultaneously. **Both of the above two special versions have stopped being supported.**

4.1 HTML/CSS

Generally, if you use WenJin Mincho in the form of WebFont on an HTML webpage, you can add the following code to CSS (taking the OTF version as an example):

```
1 .wjmin {
2     font-family: WenJinMinchoP0-Regular,WenJinMinchoP2-Regular,WenJinMinchoP3-Regular;
3 }
4 @font-face {
5     font-family: WenJinMinchoP0-Regular;
6     src: url(WenJinMinchoP0-Regular.otf)format("opentype");
7 }
8 @font-face {
9     font-family: WenJinMinchoP2-Regular;
10    src: url(WenJinMinchoP2-Regular.otf)format("opentype");
11 }
12 @font-face {
13     font-family: WenJinMinchoP3-Regular;
14     src: url(WenJinMinchoP3-Regular.otf)format("opentype");
15 }
```

Webfont technology refers to fonts that can be automatically downloaded and temporarily installed on the network to display content (such as HTML/CSS, SVG, MathML, etc.) without the user having to download and install them separately into the operating system. Developers only need to make relevant declarations in the CSS file to call Webfont.

💡 Tip

WenJin Mincho has been uploaded to [ZeoSeven Fonts Webfont CDN Service](#). Users and devel-

opers can embed the following code to call it (some functions other than Han ideographs may not work properly):

```
1 @import url('https://static.zeoseven.com/zsft/376/main/result.css');
2 @import url('https://static.zeoseven.com/zsft/377/main/result.css');
3 @import url('https://static.zeoseven.com/zsft/378/main/result.css');
4 html {
5     font-family: "WenJin Mincho Plane 0", "WenJin Mincho Plane 2", "WenJin Mincho Plane
        3";
6 }
```

For more instructions, please [click here \(in Chinese\)](#).

¶1 General OpenType Features

Generally, if you use OpenType features on an HTML webpage, you can directly use the four-character code of the OpenType feature. The following code defines the style `gb-scripttag`, using the GB/T 22321.1–2025 standard glyphs, and displaying *a* and *g* in handwriting style (aka enabling `cv02` and `ss01` features):

```
1 .gb-scripttag {
2     font-feature-settings: "cv02", "ss01";
3     -moz-font-feature-settings: "cv02", "ss01";
4     -webkit-font-feature-settings: "cv02", "ss01";
5 }
```

In HTML, using the `wjmin gb-scripttag` style can achieve the expected effect.

In addition, some OpenType features have more friendly statement names in CSS, [click here to view](#).

¶2 General Vertical Text

Inserting the following statement into the CSS style code can achieve vertical display of a section of content.

```
1 writing-mode: vertical-rl;
2 -webkit-writing-mode: vertical-rl;
3 -moz-writing-mode: vertical-rl;
```

¶3 Punctuation Kerning

Starting from Chrome/Chromium/Edge 123.0, the `text-spacing-trim` property is supported in CSS to adjust the punctuation kerning status. This property supports 6 values:

normal (Default) Compress within the paragraph; do not compress at the beginning of the paragraph, the beginning of the line within the paragraph, or the end of the line (equivalent to enabling `chws/vchw` features);

space-all Do not compress within the paragraph, at the beginning of the paragraph, the beginning of the line within the paragraph, or the end of the line (equivalent to completely disabling this feature);

space-first Compress within the paragraph and at the beginning of the line within the paragraph; do not compress at the beginning of the paragraph or the end of the line;

trim-start Compress within the paragraph, at the beginning of the paragraph, and at the beginning of the line within the paragraph; do not compress at the end of the line;

trim-both Compress within the paragraph, at the beginning of the paragraph, at the beginning of the line within the paragraph, and at the end of the line;

trim-all Set all punctuation marks to half-width form (equivalent to enabling `halt/vhal` features).

Firefox and Safari do not yet support the `text-spacing-trim` property. If you need to enable it, you need to enable it through the four-character code of the OpenType feature.

⚠ Warning

In some vertical layout environments, due to the automatic 90° rotation characteristic of some non-CJK characters, some punctuation marks may not be kerned correctly. Users can choose the following settings (non-CJK text display will be abnormal under fonts without pre-rotated non-CJK glyphs):

```
1 .vert-vrt2 {
2     writing-mode: vertical-rl;
3     -webkit-writing-mode: vertical-rl;
4     -moz-writing-mode: vertical-rl;
5     text-orientation: upright;
6     font-feature-settings: "vrt2", "liga";
7     -moz-font-feature-settings: "vrt2", "liga";
8     -webkit-font-feature-settings: "vrt2", "liga";
9 }
```

¶4 Zhuyin Vertical Layout

The following is sample CSS code for achieving Zhuyin vertical layout:

```
1 /* Use vertical Zhuyin in normal vertical layout environment */
2 .vert-upright {
3     writing-mode: vertical-rl;
4     -webkit-writing-mode: vertical-rl;
5     -moz-writing-mode: vertical-rl;
6     text-orientation: upright;
7 }
8
9 /* Environment for Zhuyin on the right of Han ideograph (vertical only), see above for
   WenJin Mincho calling */
10 .vert-std {
11     writing-mode: vertical-rl;
12     -webkit-writing-mode: vertical-rl;
13     -moz-writing-mode: vertical-rl;
14 }
15
```

```

16 .vert-std rt {
17     font-size: 30%;
18     letter-spacing: 0;
19     ruby-align: center;
20     text-align: center;
21     text-orientation: upright;
22     letter-spacing: 0;
23 }

```

Inputting Zhuyin in the `vert-upright` style allows for vertical display of Zhuyin; while using the `<ruby>` element in the `vert-std` style allows for Zhuyin annotation on the right side of Han ideograph.

4.2 L^AT_EX

4.2.1 General Settings

¶1 General OpenType Features

In X_YL^AT_EX or Lua^AT_EX, using the `fontspec` package allows the use of external fonts and OpenType features.

```

1 % Import package
2 \usepackage{fontspec}
3 \usepackage{luatexja-fontspec} % Add this line if using LuaTeX-ja package or related
   templates
4
5 % Define font calling commands
6 % (RawFeature={+xxxx,...} represents using OpenType features, using four-character codes.
   Ignore if maintaining default)
7 % [Besides the method shown above, you can also use more readable methods to call OpenType
   features (e.g., calling italic style: Style=Italic). See fontspec package documentation
   for details]
8 % (Language=xxx represents specifying language. Must be specified if using language-specific
   OpenType features. Ignore if maintaining default. See appendix for language list)
9 \newfontfamily\wjmin{WenJinMinchoP0-Regular.otf}[RawFeature={+xxxx,...},Language=xxx,...] %
   If xeCJK/LuaTeX-ja package or its template is already imported, this line is only
   effective for non-CJK text
10 \newCJKfontfamily\wjminCJK{WenJinMinchoP0-Regular.otf}[RawFeature={+xxxx,...},Language=xxx
   ,...] % xeCJK(CTeX)
11 \newjfontfamily\wjminJ{WenJinMinchoP0-Regular.otf}[RawFeature={+xxxx,...},Language=xxx,...]
   % LuaTeX-ja
12
13 % Temporarily use font
14 \fontspec{WenJinMinchoP0-Regular.otf}[RawFeature={+xxxx,...},Language=xxx,...] % If xeCJK/
   LuaTeX-ja package or its template is already imported, this line is only effective for
   non-CJK text

```

```

15 \CJKfontspec{WenJinMinchoP0-Regular.otf}[RawFeature={+xxxx,...},Language=xxx,...] % xeCJK(
    CTeX)
16 \jfontspec{WenJinMinchoP0-Regular.otf}[RawFeature={+xxxx,...},Language=xxx,...] % LuaTeX-ja
17
18 % Temporarily enable/disable features
19 % To disable a feature, change the "+" sign before the four-character code to "-"
20 \addfontfeatures{RawFeature={+xxxx,...},...}
21 \addCJKfontfeatures{RawFeature={+xxxx,...},...}
22 \addjfontfeatures{RawFeature={+xxxx,...},...}

```

¶2 Punctuation Kerning

If the xeCJK/Lua_TE_X-ja package or a template provided by it has been imported, it is recommended to use the package's own punctuation kerning settings (see the corresponding package documentation for details); otherwise, use the method above to add OpenType feature code to implement it.

¶3 Convenient Use of Italics

Utilizing the font definition function of the fontspec package allows for more convenient use of the italic form included in WenJin Mincho.

```

1 % Define font calling commands
2 % (ItalicFont specifies the font used in italic state, which can also be different features
   of the same font)
3 \newfontfamily\wjmin{WenJinMinchoP0-Regular.otf}[
4   ItalicFont=WenJinMinchoP0-Regular.otf,
5   ItalicFeatures={RawFeature={+ital}},
6   RawFeature={+xxxx,...},
7   Language=xxx
8 ] % If xeCJK/LuaTeX-ja package or its template is already imported, this line is only
   effective for non-CJK text
9 \newCJKfontfamily\wjminCJK{WenJinMinchoP0-Regular.otf}[
10  ItalicFont=WenJinMinchoP0-Regular.otf,
11  ItalicFeatures={RawFeature={+ital}},
12  RawFeature={+xxxx,...},
13  Language=xxx
14 ] % xeCJK(CTeX)
15 \newjfontfamily\wjminJ{WenJinMinchoP0-Regular.otf}[
16  ItalicFont=WenJinMinchoP0-Regular.otf,
17  ItalicFeatures={RawFeature={+ital}},
18  RawFeature={+xxxx,...},
19  Language=xxx
20 ] % LuaTeX-ja

```

4.2.2 Xe_{La}TeX

If the `xeCJK` package or a template provided by it has been imported, due to the Chinese processing mechanism of `xeCJK`, the OpenType features of the font *may* be affected. If this happens, try adding the command `\xeCJKsetup{xeCJKactive=false}` at the affected place to temporarily disable the Chinese processing function of `xeCJK`.

¶1 Automatic Switching of Plane Fonts

If the `xeCJK` package or a template provided by it has been imported, add the following code to the preamble:

```
1 \xeCJKDeclareSubCJKBlock{SIP}{"20000->"2FFFF}
2 \xeCJKDeclareSubCJKBlock{TIP}{"30000->"3FFFF}
3 \newCJKfontfamily\cjkgfont[SIP=WenJinMinchoP2-Regular,TIP=WenJinMinchoP3-Regular]{
  WenJinMinchoP0-Regular}
```

Using `\cjkgfont` allows for free switching between fonts of different planes without manual switching. In addition, `\setCJKmainfont` can be used to make similar settings for the body font.

¶2 Vertical Layout of General Text and Zhuyin

Xe_{La}TeX does not support vertical layout very well and needs to be adjusted according to the situation. In a vertical layout environment, to achieve better results, use the following commands to import fonts:

```
1 \newfontfamily\wjminv{WenJinMinchoP0-Regular.otf}[
2   RawFeature={vertical:+vert,+vkern,+vrt2},Language=Chinese Simplified]
3 \newCJKfontfamily\wjminCJKv{WenJinMinchoP0-Regular.otf}[
4   RawFeature={vertical:+vert,+vkern,+vrt2},Language=Chinese Simplified]
```

Since the Chinese processing mechanism of `xeCJK` has a significant impact on the vertical layout of Zhuyin, causing anomalies, it is recommended to add the command `\xeCJKsetup{xeCJKactive=false}` to temporarily disable the Chinese processing function of `xeCJK`.

For vertical layout reference in `xeCJK` environment: [How to output vertically typeset articles or books using LaTeX? \(in Chinese\)](#)

4.2.3 Lua_{La}TeX

¶1 Automatic Switching of Plane Fonts

Add the following code to the preamble:

```
1 \directlua{dofile("cjkgfont.lua")}
2 \newcommand{\cjkgfont}[1]{\directlua{textCJK("#1")}}
```

Similarly, fill in the following content in the `cjkgfont.lua` file (must be in the same path as the tex source file to be compiled) (under the condition that the `LuaTeX-j` package or a template provided by it has been imported):

This file is included in the `etc` folder in the GitHub repositories and can be downloaded freely.

```
1 function textCJK(s)
```

```

2 tex.sprint("\\begingroup%\n")
3 for p, c in utf8.codes(s) do
4     if ((c>=0x2E80 and c<=0x2FFF) or (c>=0x3400 and c<=0x4DFF) or
5         (c>=0x4E00 and c<=0x9FFF) or (c>=0xF900 and c<=0xFAFF)) and plane~=0 then
6         tex.sprint("\\jfontspec{WenJin Mincho Plane 0}[Renderer=HarfBuzz]")
7         plane=0
8     end
9     if c>=0x20000 and c<=0x2FFFF and plane~=2 then
10        tex.sprint("\\jfontspec{WenJin Mincho Plane 2}[Renderer=HarfBuzz]")
11        plane=2
12    end
13    if c>=0x30000 and c<=0x3FFFF and plane~=3 then
14        tex.sprint("\\jfontspec{WenJin Mincho Plane 3}[Renderer=HarfBuzz]")
15        plane=3
16    end
17    tex.sprint(utf8.char(c))
18 end
19 tex.sprint("\\endgroup%\n")
20 end

```

Please note that entering command sequences in the `\CJKfont{}` command may cause problems.

¶2 General Vertical Text

Using the vertical layout templates `ltjtarticle`/`ltjtbook`/`ltjtreport` provided by the `LuaTEX-jā` package allows for vertical layout of the entire document.

If only part of the content needs to be vertical, use the `\tate` command inside the box to switch to vertical mode.

¶3 Vertical Layout of Zhuyin

Under the condition that the `LuaTEX-jā` package or a template provided by it has been imported, add the following code to the preamble:

```

1 \ltjdefcharrange{9}{ "02C6, "02C7, "02C9, "02CA, "02CB, "02D6, "02D8, "02D9, "02EA, "02EB}
2 \newjfontfamily\wjminBpmfVert[TateFeatures={JFM={bpmfvert}}]{WenJin Mincho 0}

```

Similarly, fill in the following content in the `jfm-bpmfvert.lua` file (must be in the same path as the `tex` source file to be compiled):

This file is included in the `etc` folder in the GitHub and repositories and can be downloaded freely.

```

1 luatexja.jfont.define_jfm {
2     version = 3,
3     dir = 'tate',
4     zw = 1.0, zh = 1.0,
5     [0] = {
6         align = 'left', left = 0.0, down = 0.0,

```

```

7      width = 'prop', height = 0.88, depth = 0.12, italic = 0.0
8    },
9    [1] = { -- Zhuyin tone marks and reduced Zhuyin
10      chars = {
11        0x02C6, 0x02C7, 0x02C9, 0x02CA, 0x02CB, 0x02D6, 0x02D8, 0x02EA, 0x02EB,
12        0x0300, 0x0301, 0x0302, 0x0304, 0x0306, 0x030C, 0x1AC8, 0x31B4, 0x31B5,
13        0x31B6, 0x31B7, 0x31BB
14      },
15      align = 'left', left = 0.0, down = 0.0,
16      width = 0, height = 0.88, depth = 0.12, italic = 0.0
17    },
18    [2] = { -- Zhuyin neutral tone
19      chars = {0x02D9},
20      align = 'left', left = 0.0, down = 0.0,
21      width = 'prop', height = 0.88, depth = 0.12, italic = 0.0
22    }
23 }

```

4.3 Adobe InDesign

¶1 General OpenType Features

Most OpenType features (except cvXX) can be called directly through the OpenType settings menu in InDesign, or by selecting the character to apply the feature to and then clicking the character in the alternative character selection box that pops up next to it. If you need to use cvXX features, you need to refer to the following GitHub project:

<https://github.com/samiartur/Character-Variant-Java-Script-for-Indesign>.

You can also double-click the glyph you need in the Glyphs window to input it (open the Glyphs window from Window → Type & Tables → Glyphs).

¶2 Punctuation Kerning and Vertical Layout

InDesign itself has its own punctuation kerning and vertical layout functions. Just follow the InDesign instructions.

¶3 Vertical Layout of Zhuyin

In a vertical text frame (Adobe CJK Single-line/Paragraph Composer), input Zhuyin and select it, choose more options in the Character tab, and then enable the Rotate Roman in Vertical Text option. The tone marks will be displayed correctly. Manual adjustment may be required in some cases.

¶4 Language-Specific OpenType Features

Select a piece of text in InDesign and choose the language you want to set in the Character tab.

4.4 Other Software

4.4.1 Other Adobe Design Software (such as Photoshop, Illustrator, etc.)

¶1 General OpenType Features

Most OpenType features can be called directly through the OpenType settings menu in Adobe design software, or by selecting the character to apply the feature to and then clicking the character in the alternative character selection box that pops up next to it.

You can also double-click the glyph you need in the Glyphs window to input it (open the Glyphs window from Window → Glyphs).

¶2 Punctuation Kerning and Vertical Layout

Generally, Adobe design software itself has its own punctuation kerning and vertical layout functions. Just follow the instructions provided by Adobe.

¶3 Vertical Layout of Zhuyin

In a vertical text frame, input Zhuyin and select it. First change the language to any non-Latin language (such as Greek) in the Character tab, then select more options and enable the Standard Vertical Roman Alignment option. The tone marks will be displayed correctly. Manual adjustment may be required in some cases.

¶4 Language-Specific OpenType Features

Select a piece of text in the design software and choose the language you want to set in the Character tab.

4.4.2 Microsoft Word

¶1 OpenType Features

Microsoft Word only supports manually enabling a few features such as numbering styles (tabular/proportional figures), ligatures, and stylistic sets. Open the Font advanced settings (or Ctrl+D) → Advanced → OpenType Features, modify the settings as needed, and then click OK.

¶2 Punctuation Kerning and Vertical Layout

Microsoft Word itself has its own punctuation kerning and vertical layout functions. Just follow the Microsoft Word instructions.

¶3 Vertical Layout of Zhuyin

Microsoft Word cannot correctly position the tone marks of Zhuyin.

¶4 Language-Specific OpenType Features

Select a piece of text in Microsoft Word, click the language in the lower left corner of the editing interface, and then select the language you want to set.

4.4.3 Microsoft PowerPoint

¶1 OpenType Features

Microsoft PowerPoint cannot call OpenType features.

¶2 Punctuation Kerning

Microsoft PowerPoint does not support automatic punctuation kerning.

¶3 General Vertical Text

Microsoft PowerPoint itself has vertical layout functions for general text. Just follow the Microsoft PowerPoint instructions.

¶4 Vertical Layout of Zhuyin

Microsoft PowerPoint cannot correctly position the tone marks of Zhuyin.

¶5 Language-Specific OpenType Features

Although Microsoft PowerPoint can set the language of the text, it cannot call OpenType features specific to a particular language.

4.4.4 macOS iWork Suite (Pages, Numbers, Keynote), TextEdit, etc.

¶1 OpenType Features

Software such as the macOS iWork suite can open the Fonts dialog by clicking the screen top menu **Format** → **Font** → **Show Fonts** (or directly **Control+T**), then click the ellipsis in the upper left corner → **Typography...** to open the *Typography* dialog. Then select a piece of text and choose the font features you need in the *Typography* dialog.

¶2 Punctuation Kerning

Software such as the macOS iWork suite does not support automatic punctuation kerning.

¶3 General Vertical Text

Some software in the macOS iWork suite has vertical layout functions for general text. Just follow their instructions.

¶4 Vertical Layout of Zhuyin

Software such as the macOS iWork suite cannot correctly position the tone marks of Zhuyin.

¶5 Language-Specific OpenType Features

Software such as the macOS iWork suite cannot call OpenType features specific to a particular language.

4.5 Alternative Solutions for Applying OpenType Features in Environments that Do Not Support Them

If you need to use glyphs that can only be called through OpenType features (such as ancient book standard glyphs) in environments that do not support OpenType features (such as Excel), you can try to modify the font file using tools to achieve the desired effect.

The following tools can modify font files to enable specific OpenType features by default (please refer to their user manuals for specific usage):

⚠ Caution!

Since a few IVD glyphs share the same glyph with replacement glyphs of some OpenType features (glyph standard switching, ancient book standard characters), if you need to use the following OpenType feature freezing tools, the frozen font may cause IVD glyphs to display incorrectly. Therefore, **carefully consider using frozen fonts to display IVD glyphs.**

The ancient book standard character switching function also conflicts with the display of CJK Compatibility Ideographs, please be aware.

- OpenType Feature Freezer: <https://twardoch.github.io/fonttools-opentype-feature-freezer/>
- FontFreeze: <https://github.com/MuTsunTsai/fontfreeze>
- ttx (one of the fonttools suite): <https://github.com/fonttools/fonttools>

Use ttx to export the GSUB and cmap tables, and then modify the mapping between code points and glyphs in the cmap table. Specific modification method:

- ① Export the font's cmap table: `ttx -t cmap -o cmap.xml font.otf;`
- ② Modify the cmap.xml file with a text editor;
- ③ Merge the modified cmap table into the font file: `ttx -m cmap.xml font.otf` (add the `-b` parameter if you don't need to modify the glyph box parameters).

⚠ Warning

If the modified font file needs to be published, it must still be published under the SIL OFL 1.1 license, and the font file cannot be directly resold or peddled.

According to the requirements of the OFL license, it is strongly recommended not to use *WenJin Mincho* and its translated names anymore. Please change the name of the modified font. The purpose is to avoid confusion with the original WenJin Mincho published by the author. The minimum requirement is to change the font name to: WenJin Mincho-XX Modified Version

Chapter 5

System Fallback Font Installation Guide

5.1 Installing as a Windows System Fallback Font

⚠ Caution!

Since this operation involves certain technicalities, improper operation may cause some problems. Therefore, please follow the steps below carefully, and you also need to operate under administrator privileges.

- [1] Download the TTF version font file (standard version is recommended; choose either the three separate ttf files or the single TTF version ttc file for each version);
- [2] Select the font file in File Explorer, then right-click → Install for all users (Windows 11 users may also need to click Show more options) to install the font file;
- [3] Download the registry file reg/WenJinMincho.reg (for the C version it is reg/WenJinMinchoC.reg, for the W version it is reg/WenJinMinchoW.reg), double-click the file, then click Merge. If a security warning appears, please click Run;
- [4] Log off (sign out) and log in to the Windows system account again for the settings to take effect. To delete the font, just delete the font file itself, the registry settings do not need to be changed.
- [5] After updating the font, you still need to log off (sign out) and log in to the Windows system account again for it to take effect.
- [6] If you find that the font cannot be displayed normally (displayed as boxes) after installation/update, try restarting the system.

⚠ Warning

- This method is **only effective for the Windows GDI environment**, and ineffective for environments based on DirectWrite (such as Chromium, Firefox, WPF, UWP and other frameworks or application software based on Direct2D). There is currently no way to modify the global configuration of font fallback for this environment, because the font fallback settings of DirectWrite do not rely on the registry^①. If there is a relevant need, users can find font fallback related configurations in the application software (such as text editors) themselves, or temporarily modify it to GDI rendering (rendering effect may become worse). At the same time, developers with relevant needs are requested to configure font fallback parameters specifically for their applications;

① <https://www.unicode.org/mail-arch/unicode-ml/y2013-m07/0068.html>

- Windows 11 system cannot modify the font of Plane 3 in this way (i.e., it is always SimSun-ExtG, if the font is built-in)^❶;
- Older Windows 11 systems may fail to display Han ideographs in Extension G–J areas in some cases.

5.2 Installing as a Fallback Font for Android System (or AOSP-based Systems)

⚠ Caution!

Since this operation involves certain technicalities and risks, improper operation may cause problems or even cause the system to fail to run normally. Therefore, please follow the steps below carefully, and also ensure that your Android device has Root privileges.

- ❶ Copy the font files to the `/system/fonts/` folder;
- ❷ Backup the current Android system's `/system/etc/fonts.xml` and `/system/etc/font_fallback.xml` (if any) files;
- ❸ Open the `/system/etc/fonts.xml` and `/system/etc/font_fallback.xml` (if any) files;
- ❹ After the `<family lang="ko">` tag in the file, find the closing tag `</family>` and press Enter;
- ❺ Add the following XML code, ensuring correct indentation (choose one of the following methods, `otf/otc` version is recommended):

- a If installing three `ttf/otf` files (note that the extension in the code must correspond to the actual installed file extension):

```

1 <family>
2     <font weight="400" style="normal">WenJinMinchoP0-Regular.otf</font>
3 </family>
4 <family>
5     <font weight="400" style="normal">WenJinMinchoP2-Regular.otf</font>
6 </family>
7 <family>
8     <font weight="400" style="normal">WenJinMinchoP3-Regular.otf</font>
9 </family>
```

- b If installing a `ttc` file (note that the `ttc` type in the code must correspond to the actual installed `ttc` type):

```

1 <family>
2     <font weight="400" style="normal" index="0">WenJinMincho-OTF.ttc</font>
3 </family>
4 <family>
5     <font weight="400" style="normal" index="1">WenJinMincho-OTF.ttc</font>
6 </family>
```

❶ <https://fgwang.blogspot.com/2018/02/blog-post.html> (in Chinese)

```
7 <family>
8   <font weight="400" style="normal" index="2">WenJinMincho-OTF.ttc</font>
9 </family>
```

- 6 Restart the device to make the changes take effect.
- 7 If you still encounter problems, please check if the font files are installed correctly and ensure that the device permissions allow modification of system files.

Chapter 6

Important Notes

- Although the author strives for accuracy and ease of use when creating the font, due to limitations in technology, personal ability, and energy, there is no guarantee of the accuracy of the font data itself or that the expected rendering effect can be perfectly reproduced on your device. Therefore, the author makes no express or implied warranties regarding this font, and the author assumes no responsibility for any claims, damages, or causes attributable to your use of this font. Please understand. If you encounter problems when using this font, please provide feedback to the author as soon as possible.
- Since I am not engaged in research related to Han ideographs and do not have relevant qualifications, coupled with my limited energy and ability, I cannot guarantee that every Han ideograph glyph included in the font conforms to etymology. At the same time, Han ideographs without a G source are *in principle* not analogized. Users with relevant needs should consider carefully.
- Some IVS glyphs with different encodings are almost identical graphically. To reduce the file size and glyph count of this font^❶, these encodings will be mapped to the same glyph in the font file. Minor glyph differences (not reaching the level of separate encoding in IVS) will be accepted. Please be aware.

❶ A single sfnt format font file (ttf/otf file, not ttc/otc) can hold a maximum of 65,535 glyphs.

Chapter 7

Font License

This font is released under the [SIL Open Font License 1.1](#). **By using, copying, modifying, or distributing this font for any purpose, you acknowledge that you have fully read, understood, and agreed to the terms of the SIL Open Font License, without any objections.** Your authorization to use, modify, and distribute this font will automatically terminate if you fail to comply with any of the terms outlined in the *SIL Open Font License*.

7.1 You are permitted to:

- Use the font for any commercial or non-commercial purposes (such as typesetting, printing, design, etc.) free of charge, without the need to pay fees, notify the author, credit the source, or seek prior permission;
- Freely distribute the font files, or embed/bundle them into any software or device, even if that software or device is distributed or sold commercially;
- Modify, expand, and convert the format of the font files. However, if you distribute or publish a modified derivative font work, **the following conditions must be met:**
 - **The derivative work must remain under the SIL Open Font License 1.1**, and no additional licensing restrictions may be added;
 - **You must not use the Reserved Font Names (RFNs) or their translations in other languages: WenJin Mincho, 文津宋体, 文津宋體, 文津明朝, 문진 (文津) 명조.** This means that if you modify the glyphs, kerning, or significantly remove characters and release it as a standalone font, you must rename your derivative font;
 - **WebFont Subsetting Exception:** As a special exception, if you subset or convert the format of this font (e.g., to WOFF/WOFF2) solely for the purpose of web font delivery and optimization, and do not distribute the modified version as a downloadable, installable desktop font, you are permitted to retain the Reserved Font Names;
 - Retain the original copyright notice found in the provided license file (LICENSE.md), regardless of the modifications made.

7.2 You are strictly prohibited from:

- Selling the font files or their derivative versions **by themselves** (though they may be sold when bundled with other software);
- Releasing the font files or any modified derivative fonts under any license other than the *SIL Open Font License*;

- Using the original author's name or the Reserved Font Names to promote, endorse, or advertise your derivative font work without explicit written permission.

7.3 Disclaimers and Declarations

- The author's failure to exercise, delayed exercise, or partial exercise of any legal rights under the *SIL Open Font License* shall not be construed as a waiver of such rights, nor shall it preclude the author from exercising them in the future;
- **This is a free-for-commercial-use font. Any request for payment to acquire the font files themselves, or any requirement to purchase special permissions to use the font, is a scam. Please be cautious;**
- **Disclaimer of Warranty:** THE FONT SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. In no event shall the author be liable for any claim, damages, or other liability arising from, out of, or in connection with the font software or the use or other dealings in the font software;
- This font contains no political intent, metaphor, or purpose. Any text content generated by users typesetting with this font, any system security issues caused, or any violations of local laws and regulations are solely the personal actions of the user. The author assumes no liability whatsoever.

Chapter 8

Frequently Asked Questions

If you encounter problems when using WenJin Mincho, it is recommended to read this section first.

The content of this article is basically all taken from users' private messages and questions in the comment sections of relevant Zhihu articles. If you do not wish to have one or more of *your questions* displayed on this page, please contact the author to have them removed.

? Are there any plans to develop bold or italic versions in the future? Are there any plans to release Sans-serif (Heiti) / Imitation Song (Fangsong) / Regular Script (Kaiti) fonts in the future?

There are no plans to release the above-mentioned derivative products. Developing other large character set fonts requires a huge amount of time and energy. Among them, there are already many free commercial/open-source fonts available for Sans-serif (Heiti) large character sets, and Regular Script (Kaiti) / Imitation Song (Fangsong) large character sets are also developed by manufacturers.

For bold, it is recommended to choose [WenYuan Serif SC](#), which supports some extended Han ideographs (about 2,700). Creating bold Han ideographs through the Kage engine requires a lot of manual adjustment to reduce the phenomenon of strokes sticking together. WenJin Mincho already has built-in italics for Latin letters, which can be called via the OpenType *ital* feature. There is almost no need to develop italics for Han ideographs (if developed, they would also be oblique).

? Why does the font fallback still fail in some interfaces (Han ideographs are not displayed or cannot fallback to WenJin Mincho) after installing the registry file for setting fallback fonts?

The registry-based font fallback method is only effective for the Windows GDI environment, and is invalid for DirectWrite-based environments. For specific reasons and solutions, see 5.1 (page 37).

? Why do some punctuation marks (such as quotation marks “””, some mathematical symbols $\pm$, $\times$, $\div$, em dashes —) appear in English mode (variable width) in some application software? In software such as Word 2010 (or Word 2003, 2007), the left quotation mark leans against the left side of the Han ideograph frame, and there is a gap with the following Han ideograph?

Because the Uniscribe shaping engine equipped with the Windows GDI interface defaults to setting a piece of text (even if the text is all Chinese paragraphs) as Latin script, and WenJin Mincho supports

the automatic conversion function of punctuation forms in the non-CJK environment (default applies to Chinese full-width forms → *applies to non-CJK variable-width forms*), so in this environment, some punctuation marks will automatically convert to non-CJK styles.

Some applications (such as Word 2010, BabelStone Pad) use the GDI interface, so in the GDI environment, quotation marks will be displayed in non-CJK forms. In Word 2010, this will cause the left quotation mark to be located on the left side, leaving a gap.

Text based on [DirectWrite](#), HarfBuzz environments does not have this problem (such as Word 2013 and later versions, Google Chrome, Linux graphical interfaces).

? Why does the condensed line-spacing version have severe clipping in some interfaces? Some lowercase letters (g, j, p, q, y) are not fully displayed at the bottom?

The condensed line-spacing version itself is a special design for applications like Word. The line spacing parameters of the original version appear too large in Word, so special settings were made. Due to parameter reasons, this version will appear clipped on some interfaces. Users are advised to use the standard version on these interfaces, or modify the line spacing parameters themselves.

The author has calculated the top and bottom positions of some letters and found that: if the line spacing parameters are set to include the top and bottom content of the letters, the line spacing in Word will be expanded (in some font sizes, the behavior is different from Zhongyi Song / SimSun), which will lose the meaning of the condensed line-spacing version.

? The Bopomofo symbols in Mainland China follow the old standard, for example “一”, vertical for horizontal writing, horizontal for vertical writing. Can this be considered? I see the font example above uses horizontal for horizontal writing.

The default form of the Bopomofo symbol *i* in WenJin Mincho is a horizontal stroke. Most UI fonts (Microsoft YaHei, PingFang SC, Harmony OS Sans SC, MiSans, vivo Sans) are a horizontal stroke. You can enable the form of a vertical stroke for horizontal writing through the OpenType `hist/ss04` features.

? Why do the strokes of some extended characters still look very much like Hanazono Mincho?

Because most extended characters are generated using a modified Kage engine, the strokes of the Han ideographs generated by that engine are somewhat similar to the original Kage engine (the engine that generates Hanazono Mincho), but not exactly the same. The more significant changes include the use of Bézier curves for curved strokes (the original version uses straight lines to form polygons) and the strengthening of the serif effect of some strokes.

For specific changes, see: [Group:turgenev_KAGE エンジンの変更-技術的詳細 \(in Japanese\)](#)

? Why does WenJin Mincho appear to have excessive line spacing in Word (under default settings)?

This is determined by the font's own line spacing parameters. The line spacing parameters of the original font follow the parameter settings of Source Han Sans. The line spacing parameters of most Chinese fonts are greater than 1 times the height of the Han ideograph, which makes the line spacing appear larger in software like Word.

? Why is WenJin Mincho divided into three fonts by plane? Why not put all fonts in the same ttf/otf file? What does “Plane 0/2/3” in the font name mean?

Today's *single* ttf/otf format font (non-ttc collection) can only hold 65,535 glyphs, while WenJin Mincho supports over 110,000 Unicode Han ideographs and IVD variant characters. Therefore, a single ttf format file cannot hold all Han ideographs, so it must be divided into multiple fonts. We apologize for the inconvenience caused by technical limitations.

The *Plane 0/2/3* in the font name represents the characters of the corresponding Unicode plane supported by each font. Simply put, a Unicode plane represents a collection of characters whose Unicode values are within a certain range.

There is now a concept of a single ttf/otf format font with more than 65,535 glyphs, but it has received almost no software support. For details, please refer to: <https://github.com/harfbuzz/boring-expansion-spec/blob/main/beyond-64k.md>

? Why is WenJin Mincho an integrated version of Source Han Serif and Kage generated Han ideographs (GlyphWiki)?

Creating a font set with full Han ideograph collection requires a huge amount of time and energy. Even if it is based on relatively mature open-source body text fonts (such as Source Han Sans, Source Han Serif), it still requires making about 70,000 Han ideographs by oneself. To keep the style of extended Han ideographs consistent with existing ones is inevitably a huge project. Taking the extended Han ideograph font set based on Source Han Sans—*Plangothic* as an example, the project has continued from 2020 to the present, and under the collaborative production of multiple volunteers, it was not announced to be completed until 2025. Due to the author's limited energy, the method of integrating Source Han Serif and Kage generated Han ideographs was chosen.

At the same time, the G-source extension area Han ideographs in the GlyphWiki database are also incomplete, and the quality of existing G-source Han ideographs varies. Through component replacement, manual adjustment and other means, the author spent many days to make the Han ideographs generated by Kage meet the quantity requirements. Due to reasons of energy and level, the author did not make a large number of pseudo G-source Han ideographs for Han ideographs that do not have a G-source.

To this day, there are relatively few free commercial or even open-source large character set Han

ideograph font libraries. If you find a large character set Songti style font set other than WenJin Mincho that *simultaneously meets* the following conditions (at least the first three), please tell the author as soon as possible, thank you.

- OFL or similar font licenses that allow free commercial use and modification (such as IPA, CC0 agreement);
- Extended Han ideographs (Extension B-J) glyphs are mainly G-source (at least according to Unicode submission source information, should be G as much as possible);
- Comprehensive support for Unicode collected Han ideographs and IVD variant characters, and all are Songti style;
- Support for multiple Pinyin and Zhuyin systems;
- Existence of PostScript (Bézier cubic) curve version.

If you intend to participate in the development of WenJin Mincho, please contact the author.

? Some glyphs in this font use GlyphWiki data, so how are the glyphs generated?

Glyphs generated using GlyphWiki data are generated using a [modified version of the Kage engine](#).

The general steps are as follows:

- 1 Download [GlyphWiki glyph database](#) and unzip;
- 2 Use local Node.js to run the Kage engine (written in JavaScript) to generate vector images of glyphs (SVG format);
- 3 Use font editing software to batch import SVG vector images of glyphs.

? Has this font corrected glyphs according to future Unicode versions?

After the new version of Unicode is *officially released* (not Alpha/Beta Review stage), the author will modify the glyphs according to the latest version of the Unicode standard character table and release an updated version of this font.

? Is it suggested that this font include Suzhou numerals, circular *Fu Lu Shou Xi* symbols, Chinese chess and other symbols related to the Pan-Han character culture category?

Whether to add these symbols to this font (WenJin Mincho) is open to discussion. Personally, I think these are non-Han character symbols, and adding them is of little significance. If there is indeed a need in these aspects, you can contact the author separately for further discussion.

WenJin Mincho already includes Suzhou numerals and circular *Fu Lu Shou Xi* symbols, while the SuperHan series does not include the above symbols.

Appendix A

SIL Open Font License

Copyright (c) 2024–2026, Takushun Wu (<https://github.com/takushun-wu/>), with Reserved Font Name ‘WenJin Mincho’, ‘文津宋体’, ‘文津宋體’, ‘文津明朝’, ‘문진 (文津) 명조’.

Copyright 2010–2025 Adobe (<http://www.adobe.com/>), with Reserved Font Name ‘Source’. All Rights Reserved. Source is a trademark of Adobe in the United States and/or other countries.

Copyright 2021–2025 Tamcy (<https://github.com/chiron-fonts/chiron-sung-hk>).

Copyright 2022–2025 Shanggu Fonts (<https://github.com/GuiWonder/Shanggu>).

© 2007–2024 Adobe, But Ko, CMEX, Creative Commons Corporation, GlyphWiki & Night Koo.

Copyright 2022 The Noto Project Authors (<https://github.com/notofonts/>). Noto is a trademark of Google Inc.

Copyright © 2015 Google Inc.

Saudi Riyal Font © Emran Alhaddad - Used under SIL Open Font License 1.1

This Font Software is licensed under the SIL Open Font License, Version 1.1.

This license is copied below, and is also available with a FAQ at:

<https://openfontlicense.org>

SIL OPEN FONT LICENSE Version 1.1 - 26 February 2007

PREAMBLE

The goals of the Open Font License (OFL) are to stimulate worldwide development of collaborative font projects, to support the font creation efforts of academic and linguistic communities, and to provide a free and open framework in which fonts may be shared and improved in partnership with others.

The OFL allows the licensed fonts to be used, studied, modified and redistributed freely as long as they are not sold by themselves. The fonts, including any derivative works, can be bundled, embedded, redistributed and/or sold with any software provided that any reserved names are not used by derivative works. The fonts and derivatives, however, cannot be released under any other type of license. The requirement for fonts to remain under this license does not apply to any document created using the fonts or their derivatives.

DEFINITIONS

“Font Software” refers to the set of files released by the Copyright Holder(s) under this license and clearly marked as such. This may include source files, build scripts and documentation.

“Reserved Font Name” refers to any names specified as such after the copyright statement(s).

“Original Version” refers to the collection of Font Software components as distributed by the Copyright Holder(s).

“Modified Version” refers to any derivative made by adding to, deleting, or substituting — in part or in whole — any of the components of the Original Version, by changing formats or by porting the Font Software to a new environment.

“Author” refers to any designer, engineer, programmer, technical writer or other person who contributed to the Font Software.

PERMISSION & CONDITIONS

Permission is hereby granted, free of charge, to any person obtaining a copy of the Font Software, to use, study, copy, merge, embed, modify, redistribute, and sell modified and unmodified copies of the Font Software, subject to the following conditions:

1. Neither the Font Software nor any of its individual components, in Original or Modified Versions, may be sold by itself.
2. Original or Modified Versions of the Font Software may be bundled, redistributed and/or sold with any software, provided that each copy contains the above copyright notice and this license. These can be included either as stand-alone text files, human-readable headers or in the appropriate machine-readable metadata fields within text or binary files as long as those fields can be easily viewed by the user.
3. No Modified Version of the Font Software may use the Reserved Font Name(s) unless explicit written permission is granted by the corresponding Copyright Holder. This restriction only applies to the primary font name as presented to the users.
4. The name(s) of the Copyright Holder(s) or the Author(s) of the Font Software shall not be used to promote, endorse or advertise any Modified Version, except to acknowledge the contribution(s) of the Copyright Holder(s) and the Author(s) or with their explicit written permission.
5. The Font Software, modified or unmodified, in part or in whole, must be distributed entirely under this license, and must not be distributed under any other license. The requirement for fonts to remain under this license does not apply to any document created using the Font Software.

TERMINATION

This license becomes null and void if any of the above conditions are not met.

DISCLAIMER

THE FONT SOFTWARE IS PROVIDED “AS IS”, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT OF COPYRIGHT, PATENT, TRADEMARK, OR OTHER RIGHT. IN NO EVENT SHALL THE COPYRIGHT HOLDER BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, INCLUDING ANY GENERAL, SPECIAL, INDIRECT, INCIDENTAL, OR CONSEQUENTIAL DAMAGES, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF THE USE OR INABILITY TO USE THE FONT SOFTWARE OR FROM OTHER DEALINGS IN THE FONT SOFTWARE.

Appendix B

Donation Support

打赏支持 · Sponsor me

如果你觉得我的作品对你很有帮助的话，就多多支持我鸭~
Please sponsor me if you find my works helpful~



微信支付 · WeChat Pay
账户昵称: takushun



支付宝 · Alipay
账户昵称: takushun



Ko-fi
ko-fi.com/takushunwu

此图承载唯一正式受捐之二维码/链接信息，**请仔细甄别，谨防诈骗**
捐助与否不影响本人旗下开源项目
(<https://github.com/takushun-wu/>) 的正常使用

Precautions before sponsors and donations

- Please check the payment account carefully before giving a reward (Alipay & WeChat account name: **takushun**, if there is any change, it will be corrected at the first time). We cannot bear the loss of your fraud.
- The above payment account is the **ONLY OFFICIAL** donation account. If you find that the account information on this page does not match the payment account provided by the QR code/link, please report it immediately and assist relevant judicial authorities if necessary.
- If you are willing to donate after giving a reward, please leave a message stating your donation time and amount. Thank you for your support. Meanwhile, as a donor, you have the right to choose whether to include your information (such as nickname, personal space homepage, etc.) on public pages.
- **Once the donation amount is transferred, IT WILL NOT BE REFUNDED. Please carefully consider your decision. (Minors must obtain permission from their legal guardians before donating.)** The right to interpret the use of tipping donation fees belongs to the author himself (takushun), and **NO FORM OF SUPERVISION IS ACCEPTED.**